BSI Standards Publication

Geometrical product specifications (GPS) — Geometrical tolerancing — Maximum material requirement (MMR), least material requirement (LMR) and reciprocity requirement (RPR)

National foreword

This British Standard is the UK implementation of EN ISO 2692:2021. It is identical to ISO 2692:2021. It supersedes BS EN ISO 2692:2014, which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee TPR/1, Technical Product Realization.

A list of organizations represented on this committee can be obtained on request to its committee manager.

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Geometrical product specifications (GPS) - Geometrical tolerancing - Maximum material requirement (MMR), least material requirement (LMR) and reciprocity requirement (RPR) (ISO 2692:2021)

Spécification géométrique des produits (GPS) - Tolérancement géométrique - Exigence du maximum de matière (MMR), exigence du minimum de matière (LMR) et exigence de réciprocité (RPR) (ISO 2692:2021)

Geometrische Produktspezifikation (GPS) - Geometrische Tolerierung - Maximum-Material-Bedingung (MMR), Minimum-Material-Bedingung (LMR) und Reziprozitätsbedingung (RPR) (ISO 2692:2021)

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European foreword

This document (EN ISO 2692:2021) has been prepared by Technical Committee ISO/TC 213 "Dimensional and geometrical product specifications and verification" in collaboration with Technical Committee CEN/TC 290 “Dimensional and geometrical product specification and verification” the secretariat of which is held by AFNOR.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by December 2021, and conflicting national standards shall be withdrawn at the latest by December 2021.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

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Endorsement notice

The text of ISO 2692:2021 has been approved by CEN as EN ISO 2692:2021 without any modification.

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

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For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 213, *Dimensional and geometrical product specifications and verification*, in collaboration with the European Committee for Standardization (CEN) Technical Committee CEN/TC 290, *Dimensional and geometrical product specification and verification*, in accordance with the Agreement on technical cooperation between ISO and CEN (Vienna Agreement).

This fourth edition cancels and replaces the third edition (ISO 2692:2014), which has been technically revised.

The main changes to the previous edition are as follows:

— direct indication of maximum material or least material virtual size has been added (see 4.1.3);

— the use of SZ or CZ symbols has been added (see 4.1.4);

— the use of SIM symbol has been added (see 4.1.5).

Any feedback or questions on this document should be directed to the user’s national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

Introduction

0.1 General

This document is a geometrical product specification (GPS) standard and is to be regarded as a general GPS standard (see ISO 14638). It influences the chain links A, B and C of the chain of standards on size, form, orientation and location.

The ISO/GPS matrix model given in ISO 14638 gives an overview of the ISO/GPS system of which this document is a part. The fundamental rules of ISO/GPS given in ISO 8015 apply to this document and the default decision rules given in ISO 14253-1 apply to specifications made in accordance with this document, unless otherwise indicated.

For more detailed information on the relation of this document to the GPS matrix model, see Annex E.

This document deals with some frequently occurring workpiece functional cases in design and tolerancing. The "maximum material requirement" (MMR) can cover, for example, "assemblability" and the "least material requirement" (LMR) can cover, for example, "minimum wall thickness" of a part. MMR and LMR requirements can accurately simulate the intended function of the workpiece by allowing the combination of two independent requirements into one collective requirement or to directly define maximum material virtual condition (MMVC) or least material virtual condition (LMVC) (see Annex C). In some cases of both MMR and LMR, the "reciprocity requirement" (RPR) can be added.

NOTE 1 In GPS standards, threaded features are often considered as a type of cylindrical feature of size. However, no rules are defined in this document for how to apply MMR, LMR and RPR to threaded features. Consequently, application of the tools defined in this document for threaded features is risky.

NOTE 2 A geometrical tolerance value of 0 (0Ⓜ or 0Ⓛ) can be used to avoid non-conformity of parts that can be assembled, in the case of MMR, or have minimum wall thickness, in the case of LMR.

0.2 Information about MMR

The assembly of parts depends on the combined effect of:

a) the size (of one or more features of size), and

b) the geometrical deviation of the features and their derived features, such as the pattern of bolt holes in two flanges and the bolts securing them.

The minimum assembly clearance occurs when each of the mating features of size is at its maximum material size (MMS) (e.g. the largest bolt size and the smallest hole size) and when the geometrical deviations (e.g. the form, orientation and location deviations) of the features of size and their derived features (median line or median surface) are also fully consuming their tolerances. Assembly clearance increases to a maximum when the sizes of the assembled features of size are furthest from their MMSs (e.g. the smallest shaft size and the largest hole size) and when the geometrical deviations (e.g. the form, orientation and location deviations) of the features of size and their derived features are zero. It therefore follows that to manage the assemblability, the effect of the dimensional and geometrical variation can be dealt with by a specification using the maximum material concept. This requirement is indicated on the drawing by the symbol Ⓜ.

Furthermore, it can be useful to add Ⓜ to the datum indicator in the datum section when the datum is a feature of linear size and the clearance between the datum and the counterpart is favourable to the assembly of the part.

0.3 Information about LMR

The LMR is designed to control, for example, the minimum wall thickness, thereby preventing burst (due to pressure in a tube), or the maximum width of a series of slots. To manage the material strength function, the effect of the dimensional and geometrical variation can be dealt with by a specification using the minimum material concept. This requirement is indicated on drawings by the symbol Ⓛ.

0.4 Information about RPR

The RPR is an additional modifier, which may be used together with the MMR or with the LMR in cases where it is permitted – taking into account the function of the toleranced feature(s) – to enlarge the size tolerance when the geometrical deviation on the actual workpiece does not take full advantage of, respectively, the MMVC or the LMVC.

The RPR is indicated on drawings by the symbol Ⓡ.

0.5 General information about terminology and figures

The terminology and tolerancing concepts in this document have been updated to conform to GPS terminology, notably that in ISO 286-1, ISO 14405-1, ISO 17450-1 and ISO 17450-3.

Geometrical product specifications (GPS) — Geometrical tolerancing — Maximum material requirement (MMR), least material requirement (LMR) and reciprocity requirement (RPR)

1 Scope

This document defines the maximum material requirement (MMR), the least material requirement (LMR) and the reciprocity requirement (RPR). These requirements can only be applied to linear features of size of cylindrical type or two parallel opposite planes type.

These requirements are often used to control specific functions of workpieces where size and geometry are interdependent, for example to fulfil the functions “assembly of parts” (for MMR) or “minimum wall thickness” (for LMR). However, the MMR and LMR can also be used to fulfil other functional design requirements.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 1101:2017, *Geometrical product specifications (GPS) — Geometrical tolerancing — Tolerances of form, orientation, location and run-out*

ISO 5458, *Geometrical product specifications (GPS) — Geometrical tolerancing — Pattern and combined geometrical specification*

ISO 5459:2011, *Geometrical product specifications (GPS) — Geometrical tolerancing — Datums and datum systems*

ISO 14405-1, *Geometrical product specifications (GPS) — Dimensional tolerancing — Part 1: Linear sizes*

ISO 17450-1:2011, *Geometrical product specifications (GPS) — General concepts — Part 1: Model for geometrical specification and verification*

ISO 17450-3, *Geometrical product specifications (GPS) — General concepts — Part 3: Toleranced features*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 5459, ISO 14405-1, ISO 17450-1 and ISO 17450-3 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1
integral feature
geometrical feature belonging to the real surface of the workpiece or to a surface model

Note 1 to entry: An integral feature is intrinsically defined, for example skin of the workpiece.

[SOURCE: ISO 17450-1:2011, 3.3.5, modified — Notes 2 and 3 to entry removed.]

3.2
feature of linear size
geometrical feature, having one or more intrinsic characteristics, only one of which may be considered as variable parameter, that additionally is a member of a “one parameter family”, and obeys the monotonic containment property for that parameter

EXAMPLE 1 A single cylindrical hole or shaft is a feature of linear size. Its linear size is its diameter.

EXAMPLE 2 Two parallel opposite plane surfaces are a feature of linear size. Their linear size is the distance between the two parallel opposite planes.

[SOURCE: ISO 17450-1:2011, 3.3.1.5.1, modified — Notes to entry removed; EXAMPLE 2 replaced.]

3.3
derived feature
geometrical feature, which does not exist physically on the real surface of the workpiece and which is not natively a nominal *integral feature* (3.1)

Note 1 to entry: A derived feature can be established from a nominal integral surface, an associated integral surface or an extracted integral surface. It is qualified respectively as a nominal derived feature, an associated derived feature or an extracted derived feature.

Note 2 to entry: The centre point, the median line and the median surface defined from one or more *integral features* (3.1) are types of derived features.

EXAMPLE 1 The median line of a cylinder is a derived feature obtained from the cylindrical surface, which is an *integral feature* (3.1). The axis of the nominal cylinder is a nominal derived feature.

EXAMPLE 2 The median surface of two parallel opposite planes is a derived feature obtained from the two parallel opposite planes, which constitute an *integral feature* (3.1). The median plane of the nominal two parallel opposite planes is a nominal derived feature.

[SOURCE: ISO 17450-1:2011, 3.3.6, modified.]

3.4
maximum material size
MMS
<external feature of linear size> value equal to the upper limit of size (ULS) or to the largest ULS in case of multiple size specifications

Note 1 to entry: An MMS can be defined for any of the size characteristics in ISO 14405-1.

Note 2 to entry: ULS is defined in ISO 14405-1.

3.5
maximum material size
MMS
<internal feature of linear size> value equal to the lower limit of size (LLS) or to the smallest LLS in case of multiple size specifications

Note 1 to entry: An MMS can be defined for any of the size characteristics in ISO 14405-1.

Note 2 to entry: LLS is defined in ISO 14405-1.

3.6
least material size
LMS
<external feature of linear size> value equal to LLS or to the smallest LLS in case of multiple size specifications

Note 1 to entry: An LMS can be defined for any of the size characteristics in ISO 14405-1.

Note 2 to entry: LLS is defined in ISO 14405-1.

3.7
least material size
LMS
<internal feature of linear size> value equal to ULS or to the largest ULS in case of multiple size specifications

Note 1 to entry: An LMS can be defined for any of the size characteristics in ISO 14405-1.

Note 2 to entry: ULS is defined in ISO 14405-1.

3.8
maximum material virtual size
MMVS
value equal to the size of the *maximum material virtual condition* (3.9)

Note 1 to entry: MMVS can be directly indicated (see 4.1.3) or calculated from the *maximum material size* (3.4, 3.5) and the geometrical tolerance (see 4.1.2)

3.9
maximum material virtual condition
MMVC
state of associated feature with size equal to *maximum material virtual size* (3.8)

Note 1 to entry: MMVC is a perfect form condition of the *feature of linear size* (3.2).

Note 2 to entry: MMVC includes an orientation constraint (in accordance with ISO 1101 and ISO 5459) of the associated feature when the geometrical specification is an orientation specification (see Figure A.3). MMVC includes a location constraint (in accordance with ISO 1101 and ISO 5459) of the associated feature when the geometrical specification is a location specification (see Figure A.4).

Note 3 to entry: See examples in Annex A.

3.10
least material virtual size
LMVS
value equal to the size of the *least material virtual condition* (3.11)

Note 1 to entry: LMVS can be directly indicated (see 4.1.3) or calculated from the *least material size* (3.6, 3.7) and the geometrical tolerance (see 4.1.2)

3.11
least material virtual condition
LMVC
state of associated feature of *least material virtual size* (3.10)

Note 1 to entry: LMVC is a perfect form condition of the *feature of linear size* (3.2).

Note 2 to entry: LMVC includes an orientation constraint (in accordance with ISO 1101 and ISO 5459) of the associated feature when the geometrical specification is an orientation specification. LMVC includes a location constraint (in accordance with ISO 1101 and ISO 5459) of the associated feature when the geometrical specification is a location specification (see Figure A.5).

Note 3 to entry: See Figures A.5, A.8, A.9, A.14, A.15.

3.12
maximum material requirement
MMR
requirement for a *feature of linear size* (3.2), defining a geometrical feature of the same type and of perfect form, with a given value for the intrinsic characteristic (dimension) equal to the *maximum material virtual size* (3.8), which limits the non-ideal feature on the outside of the material

Note 1 to entry: MMR is used to control the assemblability of a workpiece.

Note 2 to entry: See also 4.2.

3.13
least material requirement
LMR
requirement for a *feature of linear size* (3.2), defining a geometrical feature of the same type and of perfect form, with a given value for the intrinsic characteristic (dimension) equal to the *least material virtual size* (3.10), which limits the non-ideal feature on the inside of the material

Note 1 to entry: LMR is used, for example, to control the minimum wall thickness between two symmetrical or coaxially located similar features of size.

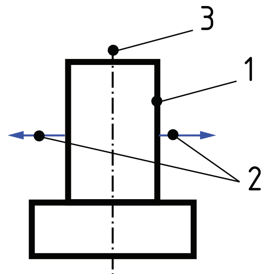
Note 2 to entry: See also 4.3.

3.14
reciprocity requirement
RPR
additional requirement for a *feature of linear size* (3.2) indicated in addition to the *maximum material requirement* (3.12) or the *least material requirement* (3.13) to indicate that the size tolerance is increased by the difference between the geometrical tolerance and the actual geometrical deviation

3.15
external feature of linear size
feature of linear size (3.2) where vectors normal to the surface are directed outward from the material in a direction opposite to the median feature

Note 1 to entry: The cylindrical surface of a shaft is considered to be an external feature of linear size.

Note 2 to entry: See Figure 1.



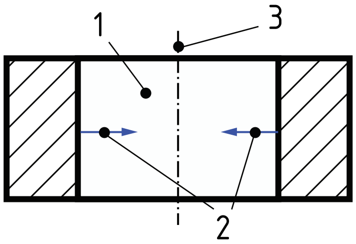
Key
1 external feature of linear size
2 normal vectors directed outward from the material
3 median feature (cylinder axis)

Figure 1 — Example of external feature of linear size

3.16
internal feature of linear size
feature of linear size (3.2) where vectors normal to the surface are directed outward from the material in a direction toward the median feature

Note 1 to entry: The cylindrical surface of a hole is considered to be an internal feature of linear size.

Note 2 to entry: See Figure 2.



Key
1 internal feature of linear size
2 normal vectors directed outward from the material
3 median feature (cylinder axis)

Figure 2 — Example of internal feature of linear size

4 Maximum material requirement (MMR) and least material requirement (LMR)

4.1 General

4.1.1 MMVS or LMVS specification

The MMR and the LMR can be applied to a set of one or more feature(s) of size as toleranced feature(s), datum(s) or both. The MMVS or the LMVS shall be specified by one of the two following options:

a) An MMR without direct indication of MMVS or an LMR without direct indication of LMVS but with a size specification for the considered feature. This option is referred to as indirect determination of virtual size in this document.

b) An MMR with direct indication of MMVS between square brackets in the tolerance indicator or an LMR with direct indication of LMVS between square brackets in the tolerance indicator as explained in this document. This option is referred to as direct indication of virtual size.

A geometrical specification with MMR or LMR shall be indicated as applying to a derived feature. However, the toleranced feature considered in the rules of this document is the corresponding integral feature.

The rules in this document shall not be applied to threaded features, even if threaded features are often considered as cylindrical features in GPS standards.

The possible combinations of geometrical characteristic symbols and MMR or LMR are illustrated in Annex D.

4.1.2 Indirect determination of MMVS or LMVS

When indirect determination of virtual size is selected [4.1.1 a)], the virtual size(s) introduced by the use of maximum or least material modifier in geometrical specification shall be calculated by considering the combination(s) of the geometrical tolerance(s) (applied to the derived feature of the feature of size) and the upper or lower tolerance limit of the dimensional specification(s) (of the feature(s) of size).

NOTE As limited by the scope, the only derived features considered in this document are median lines and median surfaces.

When indirect determination of virtual size is used, then the MMVS or the LMVS shall be the result of the computations described hereafter.

For external features of linear size except for a datum feature with MMR when rule F is fulfilled [see 4.2.2 b)], the MMVS is given by Formula (1):

$$\sigma_{\mathrm{MMVS}} = \sigma_{\mathrm{MMS}} + \delta \tag{1}$$

For internal features of linear size except for a datum feature with MMR when rule F is fulfilled [see 4.2.2 b)], the MMVS is given by Formula (2):

$$\sigma_{\mathrm{MMVS}} = \sigma_{\mathrm{MMS}} - \delta \tag{2}$$

For external features of linear size except for a datum feature with LMR when rule M is fulfilled [see 4.3.2 b)], the LMVS is given by Formula (3):

$$\sigma_{\mathrm{LMVS}} = \sigma_{\mathrm{LMS}} - \delta \tag{3}$$

For internal features of linear size except for a datum feature with LMR when rule M is fulfilled [see 4.3.2 b)], the LMVS is given by Formula (4):

$$\sigma_{\mathrm{LMVS}} = \sigma_{\mathrm{LMS}} + \delta \tag{4}$$

For a datum feature with MMR when rule F is fulfilled [see 4.2.2 b)], the MMVS for external and internal features of size is given by Formula (5):

$$\sigma_{\mathrm{MMVS}} = \sigma_{\mathrm{MMS}} \tag{5}$$

For a datum feature with LMR when rule M is fulfilled (see 4.3.2 b)], the LMVS for external and internal features of size is given by Formula (6):

$$\sigma_{\mathrm{LMVS}} = \sigma_{\mathrm{LMS}} \tag{6}$$

where

σ_{MMVS} is the MMVS;

σ_{MMS} is the MMS;

σ_{LMVS} is the LMVS;

σ_{LMS} is the LMS;

δ is the geometrical tolerance.

4.1.3 Direct indication of MMVS or LMVS

When direct indication of virtual size is selected [4.1.1 b)], then the MMVS or the LMVS shall be indicated between square brackets in the tolerance indicator and the virtual size is equal to this value as stated in the rules of this document. If a size is also specified for the considered feature, it shall be considered as an independent specification according to ISO 14405-1. No collective requirement is created between the two specifications (size specification and geometrical specification) in the case of direct indication of MMVS or LMVS.

NOTE It is the responsibility of the designer to select compatible values for the size of the feature and the MMVS or LMVS as they can conflict.

4.1.4 MMR or LMR applied to several toleranced features

When an MMR or LMR applies to several toleranced features, the symbols CZ or SZ shall always be indicated in the zone section of the tolerance indicator following the sequence order specified in ISO 1101.

NOTE See Annex B for former practice.

4.1.5 Simultaneous requirement

A simultaneous requirement can be useful for example for MMR or LMR with same datum indication containing MMR or LMR.

When a simultaneous requirement is needed, the SIM symbol possibly followed by an identification number (SIMi) without a space shall be indicated in the adjacent indication area of each related geometrical specification in accordance with ISO 5458.

The use of the SIM modifier transforms a set of more than one geometrical specification with MMR or LMR into a combined specification. The corresponding MMVC or LMVC are locked together with location and orientation constraints according to the rules of this document. The datum system is also constrained to be the same for each specification in the same SIM group.

Figure A.17 shows an example of simultaneous requirement.

4.1.6 MMR or LMR on a datum without MMR or LMR on the toleranced feature

When an MMR or LMR is applied to the datum only (see Figure A.19), then the rules for datum fully apply (see 4.2.2, 4.2.4 and 4.3.2, 4.3.4). In addition, the constraints on the MMVC(s) of the datum(s) and the MMVC(s) of the toleranced feature(s) stated in rule D [see 4.2.1 d)] or in rule K (see 4.3.1 d)] are replaced with the corresponding constraints applied on the MMVC(s) of the datum(s) and the tolerance zone as defined in ISO 1101 and ISO 5459.

4.2 Maximum material requirement (MMR)

4.2.1 MMR for toleranced features with indirect determination of virtual size

When the MMR applies to the toleranced feature and the indirect determination of virtual size is selected, it shall be indicated on drawings by the symbol Ⓜ placed after the geometrical tolerance of the derived feature of the feature of linear size (toleranced feature) in the tolerance indicator with no size indicated in square brackets.

The MMR for toleranced features with indirect determination of virtual size results in four independent requirements:

— a requirement for the upper limit of the size [see rules A 1) and A 2)];

— a requirement for the lower limit of the size [see rules B 1) and B 2)];

— a requirement for the surface non-violation of the MMVC (see rule C);

— a requirement for applying constraints on MMVCs (see rule D).

When the MMR is specified for the toleranced feature with indirect determination of virtual size, then the following rules shall be applied for the surface(s) (of the feature of linear size), and the MMVS shall be computed from the size specification and the geometric specification according to the rules of this document.

a) **Rule A:** The sizes of the toleranced feature shall be:

   1) equal to or smaller than the MMS, for external features;

   2) equal to or larger than the MMS, for internal features.

   NOTE 1 This rule can be altered by the indication of RPR, with symbol Ⓡ after the symbol Ⓜ [see 5.2 and Figure A.1 b)].

b) **Rule B:** The sizes of the toleranced feature shall be:

   1) equal to or larger than the LMS, for external features [see Figures A.2 a), A.3 a), A.4 a), A.6 a), A.7 a), A.10, A.11 and A.12];

   2) equal to or smaller than the LMS, for internal features [see Figures A.2 b), A.3 b), A.4 b), A.6 b), A.7 b), A.10 to A.13, A.16 to A.19].

c) **Rule C:** The MMVC of the toleranced feature shall not be violated by the extracted (integral) feature (see Figures A.2, A.3, A.4, A.6, A.7, A.10 to A.19).

   NOTE 2 Use of the envelope requirement Ⓔ may lead to superfluous constraints, reducing the technical and economic advantage of MMR, if the functional requirement is purely assemblability.

   NOTE 3 The indication 0Ⓜ applied to a form specification on a feature of linear size has the same meaning as the envelope requirement Ⓔ applied to a size.

d) **Rule D:** The rule applies as follows:

   — When the geometrical specification is an orientation or a location relative to a (primary) datum or a datum system, the MMVC of the toleranced feature shall be in theoretically exact orientation or location relative to the datum or the datum system, in accordance with ISO 1101 and ISO 5459 (see 3.9, Note 2 to entry, and Figures A.3, A.4, A.6 and A.7).

   — Moreover, if several toleranced features are controlled by the same tolerance indication with the CZ symbol, the MMVCs shall also be in theoretically exact orientation and location relative to each other (see Figures A.1, A.10, A.11 and A.13).

   — If several toleranced features are controlled by the same tolerance indicator with the SZ symbol, then the MMVCs are not constrained to be in theoretically exact orientation nor location relative to each other (see Figure A.18). In both cases (CZ or SZ indication) constraints relative to datums remain.

   — Additionally, if the symbol SIM possibly followed by an index number as required by ISO 5458 is indicated, then the MMVCs shall be constrained in orientation and location with the MMVCs of the SIM group.

4.2.2 MMR for related datum features with indirect determination of virtual size

When the MMR applies to the datum feature and the indirect determination of virtual size is selected, it shall be indicated on drawings by the symbol Ⓜ placed after the datum letter(s) in the tolerance indicator.

The datum letter(s) followed by the symbol Ⓜ result in an associated feature with a fixed size defined by the MMVC.

NOTE 1 Virtual conditions of toleranced feature and datum feature are constrained between them in orientation and location. The result is one combined virtual condition.

The MMR for datum features results in three independent requirements:

— a requirement for the surface non-violation of the MMVC (see rule E);

— a requirement for MMVS when there is no geometrical specification or when there are only geometrical specifications whose tolerance value is not followed by the symbol Ⓜ (see rule F);

— a requirement for MMVS when there is a geometrical specification whose tolerance value is followed by symbol Ⓜ and whose "datum" section (third and subsequent compartments) of the tolerance indicator meets a property defined in rule G.

NOTE 2 The use of Ⓜ after the datum letter is only possible if the datum is obtained from a feature of linear size.

When MMR applies to all elements of the collection of surfaces of a common datum, the corresponding sequence of letters identifying the common datum shall be indicated within parentheses (see Figure A.13 and ISO 5459:2011, rule 9) and MMVCs are by default constrained in location and orientation relative to each other (see ISO 5459:2011, rule 7). When MMR applies only to one surface of the collection of features involved in a common datum, the sequence of letters identifying the common datum shall not be indicated within parentheses, and the requirement applies only to the feature identified by the letter placed just before the modifier.

In this case, it specifies for the surface(s) (of the feature of linear size) the following rules:

a) **Rule E:** The MMVC of the related datum feature shall not be violated by the extracted (integral) datum feature from which the datum is derived (see Figures A.6 and A.7). If a SIM symbol optionally followed by an index number as required by ISO 5458 is indicated in the adjacent zone of the corresponding tolerance indicator, then the MMVC shall be in the same orientation and location as the MMVCs considered for the datums of the same SIM group (see Figure A.17).

b) **Rule F:** The MMVS of the related datum feature shall be equal to its MMS [see Formula (5)], when the related datum feature:

   — has no geometrical specification (see Figure A.6); or

   — has only geometrical specifications whose tolerance value is not followed by the symbol Ⓜ; or

   — has no geometrical specification conforming with rule G.

   NOTE 3 It is emphasized that when the related datum feature has a geometrical specification without the symbol Ⓜ, then rule F is not sufficient to ensure the assemblablity of the datum feature. See the computation of MMVS in 4.1.2 and Figure A.6 c). Additionally, if the geometrical specification of the datum feature is modified in a subsequent revision of the drawing, then the MMVS could change if rule G becomes mandatory in this case.

c) **Rule G:** The MMVS of the related datum feature shall be equal to the MMS, plus (for external features of linear size) or minus (for internal features of linear size) the geometrical tolerance, when the datum feature is controlled by a geometrical specification whose tolerance value is followed by the symbol Ⓜ, and:

   1) it is a form specification, and the related datum is the primary datum of the tolerance indicator where the symbol Ⓜ is indicated next to the datum letter (see Figure A.7); or

   2) it is an orientation or location specification whose datum or datum system contains exactly the same datum(s) in the same order as the one(s) called before the related datum in the tolerance indicator where the Ⓜ symbol is indicated next to the datum letter (see Figures A.12 and A.13).

   NOTE 4 In this case, the MMVS for external features of linear size is as given in Formula (1), and the MMVS for internal features of linear size is as given in Formula (2). See 4.1.2.

   When these properties are not observed, rule F applies.

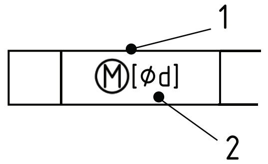
In the case of rule G, the datum feature indicator should be directly connected to that geometrical tolerance indicator from which the MMVC of the datum feature is controlled (see ISO 5459:2011, rule 1, bullet 2).

NOTE 5 This recommendation was a requirement in ISO 2692:2014 (see B.3).

When several geometrical specifications on a related datum feature fulfil rule G, it is recommended that direct indication of MMVS is used (see 4.1.3).

4.2.3 MMR for toleranced features with direct indication of virtual size

When the MMR applies to the toleranced feature and the direct indication of virtual size is selected, then the value of MMVS shall be indicated after the symbol Ⓜ between square brackets in the zone, feature and characteristic section. The value of MMVS shall be preceded by a Ø symbol for a cylinder (see Figure 3 for a cylindrical feature). The Ø symbol shall be omitted when the feature size is not a diameter. When direct indication of MMVS is selected, no geometrical tolerance value shall be indicated before the symbol Ⓜ in the zone, feature and characteristic section (see Figures A.14 and A.15).



Key

1 zone, feature and characteristic section

2 numerical value for the MMVS

Figure 3 — Direct indication of MMVS for a toleranced feature

When the MMR is specified for the toleranced feature with the direct indication of virtual size, rules A and B shall not be applied and rules C, D and Q shall be applied.

Rule Q: The MMVS for the toleranced feature shall be equal to the value indicated between square brackets in the corresponding zone, feature and characteristic section.

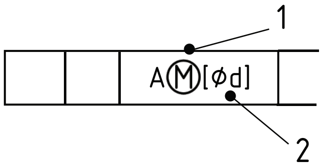
If any size is also specified for the considered feature, it shall be considered as an independent specification according to ISO 14405-1 (see Figures A.14 and A.15).

NOTE See 4.1.3.

4.2.4 MMR for related datum features with direct indication of virtual size

When the MMR applies to the datum and the direct indication of virtual size is selected, then the value of MMVS shall be indicated after the symbol Ⓜ between square brackets in the datum indicator (see Figure A.16). The value of MMVS shall be preceded by a Ø symbol for a cylinder (see Figure 4 for a cylindrical feature). The Ø symbol shall be omitted when the feature size is not a diameter.

NOTE The value indicated between square brackets is the designer's responsibility. This value can be different from the calculated value when applying the indirect determination of MMVS.



Key

1 datum section

2 numerical value for the MMVS of the datum

Figure 4 — Direct indication of MMVS for a datum

In the case of a common datum with the same MMVS for each datum feature, the indication of common datum in the datum indicator section shall be surrounded with brackets, followed by Ⓜ and the indication of the MMVS in square brackets, e.g. (A-B)Ⓜ [d].

In the case of a common datum with different MMVSs for each datum feature, the indication of common datum in the datum indicator section shall be surrounded with brackets, followed by Ⓜ and the indication of the MMVSs in square brackets separated with a dash and surrounded with brackets, e.g. (A-B)Ⓜ ([d1]-[d2]). Each MMVS value applies to each datum feature in the same order.

These two requirements apply in the same manner for more than two datum features.

When the MMR is specified for a datum with direct indication of the MMVS then rule E shall be applied, rules F and G shall not be applied and rule O shall be applied to the surface of the feature of linear size.

Rule O: The MMVS for the related datum feature shall be equal to the value indicated between square brackets in the corresponding datum section.

If any size is also specified for the considered feature, it shall be considered as an independent specification according to ISO 14405-1.

4.3 Least material requirement (LMR)

4.3.1 LMR for toleranced features with indirect determination of virtual size

When the LMR applies to the toleranced feature and the indirect determination of virtual size is selected, then it shall be indicated on the drawing by the symbol Ⓛ placed after the geometrical tolerance of the derived feature of the feature of linear size (toleranced feature) in the tolerance indicator with no size indicated in square brackets.

EXAMPLE To fully control the minimum wall thickness, the symbol Ⓛ can be applied to the tolerancing of the features on both sides of the wall. LMR can be implemented in two different ways, as follows:

— The location requirements for the two different sides of the wall can refer to the same datum axis or datum system (see Figure A.8). In this case, Ⓛ applies to the two toleranced features.

— The location requirement of the derived feature for one of the sides of the wall can refer to the derived feature of the other as the datum. In this case, the tolerance for the toleranced feature and the datum letter are followed by the symbol Ⓛ (see Figure A.9).

NOTE 1 This possibility only applies if the features on the two sides are features of size.

When the LMR is specified for the toleranced feature with indirect determination of virtual size, then the following rules shall be applied to the surface(s) (of the feature of linear size), and the LMVS shall be computed from the size specification and the geometric specification according to the rules of this document.

a) **Rule H:** The sizes of the toleranced feature shall be:

1) equal to or larger than the LMS, for external features;

2) equal to or smaller than the LMS, for internal features.

NOTE 2 This rule can be altered by the indication of RPR, with the symbol Ⓡ after the symbol Ⓛ [see 5.3, Figure A.5 e) and f)].

b) **Rule I:** The sizes of the toleranced feature shall be:

1) equal to or smaller than the MMS, for external features [see Figures A.5 a), A.8 and A.9];

2) equal to or larger than the MMS, for internal features [see Figures A.5 b) and A.8].

c) **Rule J:** The LMVC of the toleranced feature shall not be violated by the extracted (integral) feature (see Figures A.5, A.8 and A.9).

d) **Rule K:** The rule applies as follows:

— When the geometrical specification is an orientation or a location relative to a (primary) datum or a datum system, the LMVC of the toleranced feature shall be in theoretically exact orientation or location relative to the datum or the datum system, in accordance with ISO 1101 and ISO 5459 (see 3.11, Note 2 to entry, and Figures A.5, A.8 and A.9).

— Moreover, if several toleranced features are controlled by the same tolerance indication with CZ symbol, the LMVCs shall also be in theoretically exact orientation and location relative to each other.

— If several toleranced features are controlled by the same tolerance indicator with the SZ symbol, then the LMVCs are not constrained to be in theoretically exact orientation nor location relative to each other (see Figure A.18 for an illustration in the MMR case). In both cases (CZ or SZ indication), possible constraints relative to datums remain.

— Additionally, if the symbol SIM possibly followed by an index number as required by ISO 5458 is indicated then the LMVCs shall be constrained in orientation and location with the LMVCs of the SIM group.

4.3.2 LMR for related datum features with indirect determination of virtual size

When the LMR applies to the datum feature and the indirect determination of virtual size is selected, it shall be indicated on the drawing by the symbol Ⓛ placed after the datum letter(s) in the tolerance indicator.

The datum letter(s) followed by Ⓛ result in an associated feature with a fixed size defined by the LMVC.

NOTE 1 Virtual conditions of toleranced feature and datum feature are constrained between them in orientation and location. The result is one combined virtual condition.

The LMR for datum features results in three independent requirements:

— a requirement for the surface non-violation of the LMVC (see rule L);

— a requirement for LMVS when there is no geometrical specification or when there are only geometrical specifications whose tolerance value is not followed by the symbol Ⓛ (see rule M);

— a requirement for LMVS when there is a geometrical specification whose tolerance value is followed by the symbol Ⓛ and whose “datum” section (third and subsequent compartments) of the tolerance indicator meets a property defined in rule N.

NOTE 2 The use of Ⓛ after the datum letter is only possible if the datum is obtained from a feature of linear size.

When LMR applies to all elements of the collection of surfaces of a common datum, the corresponding sequence of letters identifying the common datum shall be indicated within parentheses (see Figure A.13 for an illustration of a similar case with MMR instead of LMR and ISO 5459:2011, rule 9) and LMVCs are by default constrained in location and orientation relative to each other (see ISO 5459:2011, rule 7). When LMR applies only to one surface of the collection of features involved in a common datum, the sequence of letters identifying the common datum shall not be indicated within parentheses, and the requirement applies only to the feature identified by the letter placed just before the modifier.

In this case, it specifies for the surface(s) (of the feature of linear size) the following rules.

a) **Rule L:** The LMVC of the related datum feature shall not be violated by the extracted (integral) datum feature from which the datum is derived (see Figure A.9). If a SIM symbol optionally followed by an index number as required by ISO 5458 is indicated in the adjacent zone of the corresponding indicator, then the LMVC shall be in the same orientation and location as the LMVCs considered for the datums of the same SIM group.

b) **Rule M:** The LMVS of the related datum feature shall be equal to its LMS [see Formula (6)], when the related datum feature:

— has no geometrical specification (see Figure A.9); or

— has only geometrical specifications whose tolerance value is not followed by the symbol Ⓛ; or

— has no geometrical specification conforming with rule N.

c) **Rule N:** The LMVS of the related datum feature shall be equal to the LMS, minus (for external features of linear size) or plus (for internal features of linear size) the geometrical tolerance, when the datum feature is controlled by a geometrical specification whose tolerance value is followed by the symbol Ⓛ, and:

1) it is a form specification, and the related datum corresponds to the primary datum of the tolerance indicator where the symbol Ⓛ is indicated next to the datum letter; or

2) it is an orientation or location specification whose datum or datum system contains exactly the same datum(s) in the same order as the one(s) called before the related datum in the tolerance indicator where the symbol Ⓛ is indicated next to the datum letter.

NOTE 3 In this case, the LMVS for external features of linear size is as given in Formula (3), and the LMVS for internal features of linear size is as given in Formula (4). See 4.1.2.

When these properties are not observed, rule M applies.

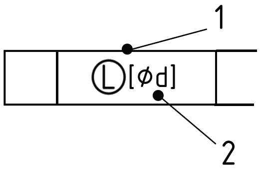
In the case of rule N, the datum feature indicator should be directly connected to that geometrical tolerance indicator from which the LMVC of the datum feature is controlled (see ISO 5459:2011, rule 1, bullet 2).

NOTE 4 This recommendation was a requirement in ISO 2692:2014 (see B.3).

When several geometrical specifications on a related datum feature fulfil rule N, it is recommended that direct indication of LMVS is used (see 4.1.3).

4.3.3 LMR for toleranced features with direct indication of virtual size

When the LMR applies to the toleranced feature and the direct indication of virtual size is selected, then the value of LMVS shall be indicated after the symbol Ⓛ between square brackets in the zone, feature and characteristics section. The value of LMVS shall be preceded by a Ø symbol for a cylinder (see Figure 5 for a cylindrical feature). The Ø symbol shall be omitted when the feature size is not a diameter. When direct indication of LMVS is selected, no geometrical tolerance value shall be indicated before the symbol Ⓛ in the zone, feature and characteristic section (see Figures A.14 and A.15).



Key

1 zone, feature and characteristic section

2 numerical value for the LMVS

Figure 5 — Direct indication of LMVS for a toleranced feature

When the LMR is specified for the toleranced feature with the direct indication of virtual size, rules H and I shall not be applied and rules J, K and R shall be applied.

Rule R: The LMVS for the toleranced feature shall be equal to the value indicated between square brackets in the corresponding zone, feature and characteristic section.

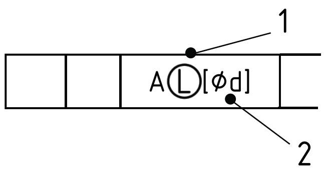
If any size is also specified for the considered feature, it shall be considered as an independent specification according to ISO 14405-1 (see Figures A.14 and A.15).

NOTE See 4.1.3.

4.3.4 LMR for related datum features with direct indication of virtual size

When the LMR applies to the datum and the direct indication of virtual size is selected, then the value of LMVS shall be indicated after the symbol Ⓛ between square brackets in the datum indicator. The value of LMVS shall be preceded by a Ø symbol for a cylinder (see Figure 6 for a cylindrical feature). The Ø symbol shall be omitted when the feature size is not a diameter.

NOTE The value indicated between square brackets is the designer's responsibility. This value can be different from the calculated value when applying the indirect determination of LMVS.



Key

1 datum section

2 numerical value for the LMVS of the datum

Figure 6 — Direct indication of LMVS of a datum

In the case of a common datum with the same LMVS for each datum feature, the indication of common datum in the datum indicator section shall be surrounded with brackets, followed by Ⓛ and the indication of the LMVS in square brackets, e.g. (A-B)Ⓛ [d].

In the case of a common datum with different LMVSs for each datum feature, the indication of common datum in the datum indicator section shall be surrounded with brackets, followed by Ⓛ and the indication of the LMVSs in square brackets separated with a dash and surrounded with brackets, e.g. (A-B)Ⓛ ([d1]-[d2]). Each LMVS value applies to each datum feature in the same order.

These two requirements apply in the same manner for more than two datum features.

When the LMR is specified for a datum with direct indication of the LMVS, then rule L shall be applied, rules M and N shall not be applied, and rule P shall be applied to the surface of the feature of linear size.

Rule P: The LMVS for the related datum feature shall be equal to the value indicated between square brackets in the corresponding datum section.

If any size is also specified for the considered feature, it shall be considered as an independent specification according to ISO 14405-1.

5 Reciprocity requirement (RPR)

5.1 General

RPR shall be indicated on drawings as an additional requirement to MMR or LMR by the symbol Ⓡ placed after the symbol Ⓜ, or by the symbol Ⓡ placed after the symbol Ⓛ, respectively. RPR shall only be indicated in the zone, feature and characteristic section of the tolerance indicator. The modifier Ⓡ shall not be indicated in the datum section of the tolerance indicator.

RPR shall be applied only with indirect determination of virtual size (see 4.1.1).

The additional requirement, RPR, alters the size tolerance of the feature of linear size in the collective requirements MMR and LMR. By means of RPR, the size can take full advantage of the MMVC and the LMVC. RPR allows for the choice of distribution of variation allowance between dimensional and geometrical tolerances based on manufacturing capabilities.

NOTE RPR can express the same workpiece functions as the indication “0Ⓜ”.

5.2 Reciprocity requirement (RPR) and maximum material requirement (MMR)

When the reciprocity symbol Ⓡ follows the symbol Ⓜ indicating an RPR, it alters the MMR for the surface(s) (of the feature of linear size) in the following ways [see Figure A.1 b)]:

— Rule A is not valid;

— Rules B to D are still valid.

NOTE The RPR allows an increase in the dimensional tolerance when the geometrical deviation does not take full advantage of the MMVC.

The reciprocity symbol defines equivalent requirements to specifications with dimensional and geometrical requirements with 0Ⓜ.

5.3 Reciprocity requirement (RPR) and least material requirement (LMR)

When the reciprocity symbol Ⓡ follows the symbol Ⓛ indicating an RPR, it alters the LMR for the surface(s) (of the feature of linear size) in the following ways [see Figure A.5 e) and f)]:

— Rule H is not valid;

— Rules I to K are still valid.

NOTE The RPR allows an increase in the dimensional tolerance when the geometrical deviation does not take full advantage of the LMVC.

The reciprocity symbol defines equivalent requirements to specifications with dimensional and geometrical requirements with 0Ⓛ.

Annex A
(informative)

Examples of tolerancing with Ⓜ, Ⓛ and Ⓡ

Figures in this document are intended only as illustrations to aid the user in understanding the MMR, the LMR and the RPR. In some instances, figures provide added details for emphasis; in other instances, figures have deliberately been left incomplete. Numerical values of dimensions and tolerances have been given for illustrative purposes only.

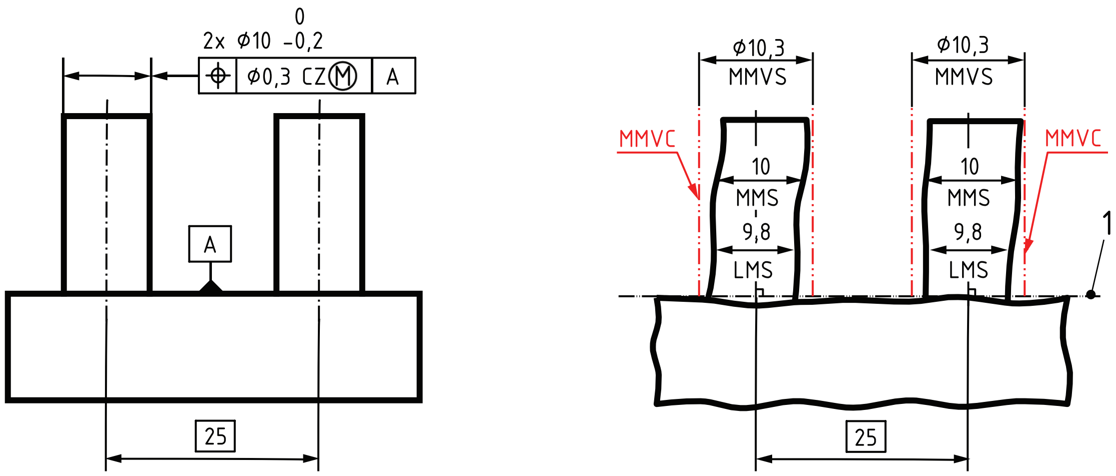
In the interpretation figures, the MMS and LMS are shown as if they were visible in the two-dimensional views presented and the dimension lines (with arrows) are attached to the line representing the real part. However, in some figures this is not possible (see Figures A.10, A.11 and A.12) and MMS or LMS are only suggested with floating dimension lines (with arrows). In every case the value of the MMS or LMS is meaningful rather than the place where it is shown. The orientation of the local sizes (two-point sizes) on the real workpiece is not necessarily perpendicular to the axis of the MMVC or LMVC.

For uniformity in this document, all dimensions are given in millimetres and all figures are in first angle projection.

It should be understood that the third angle projection method could equally well have been used to illustrate the principles established. For the definitive presentation (proportions and dimensions) of symbols for geometrical tolerancing, see ISO 7083.

In the explanation under each figure, conditions on local diameters and local sizes systematically include the limit value ("equal to" is therefore implied every time the expressions "larger than" or "smaller than" are employed).

Drawing indication | **Interpretation**



The intended function of the part illustrated in Figure A.1 a) is an assembly with a plate with two holes 25 mm apart. The holes are required to be perpendicular to the contact surface of the plate.

The interpretation is based on the following rules and definitions given in this document.

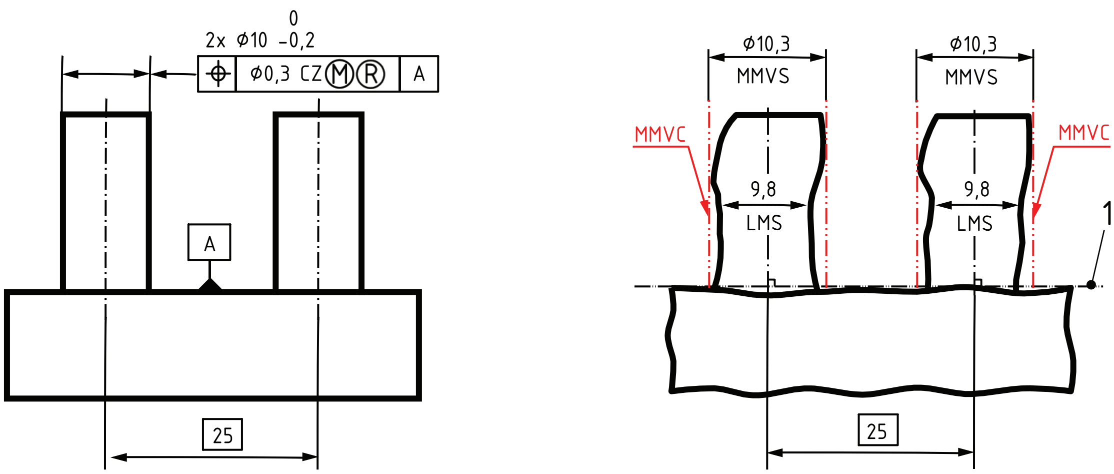
i) The extracted feature of the toleranced pins shall not violate the MMVC, which has the diameter MMVS = 10,3 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature of the toleranced pins shall have everywhere a local diameter larger than LMS = 9,8 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 10,0 mm [see rule A 1), 3.4 and 3.5].

iii) The location of the two MMVCs is theoretically exact: at a distance 25 mm relative to each other and perpendicular to the datum A [see rule D and 3.9, Note 2 to entry].

a) Example of MMR without RPR for two external cylindrical features based on size and location (position) requirements

Drawing indication **Interpretation**



The intended function of the part illustrated in Figure A.1 b) is an assembly with a plate with two holes 25 mm apart. The holes are required to be perpendicular to the contact surface of the plate.

The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced pins shall not violate the MMVC, which has the diameter MMVS = 10,3 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature of the toleranced pins shall have everywhere a local diameter larger than LMS = 9,8 mm [see rule B 1), 3.6 and 3.7]. There is no requirement concerning the upper limit of local diameters (see 5.2). The RPR requirement allows the size tolerance to increase.

iii) The location of the two MMVCs is theoretically exact: at a distance 25 mm relative to each other and perpendicular to the datum A [see rule D and 3.9, Note 2 to entry].

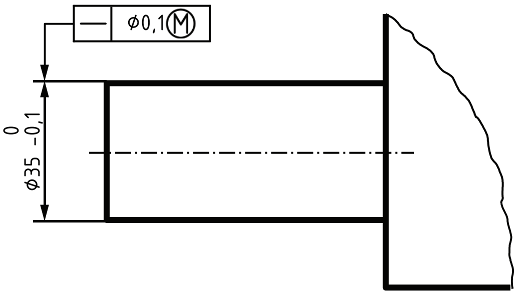
b) Example of MMR with RPR for two external cylindrical features based on size and location (position) requirements

Key

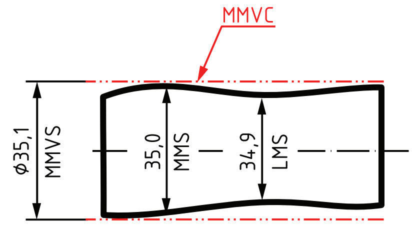
1 datum A

Figure A.1 — Examples of MMR for two external cylindrical features based on size and location (position) requirements

Drawing indication



Interpretation



The intended function of the part toleranced in Figure A.2 a) could be a clearance fit with a hole of the same length as the toleranced cylindrical feature.

The interpretation is based on the following rules and definitions given in this document.

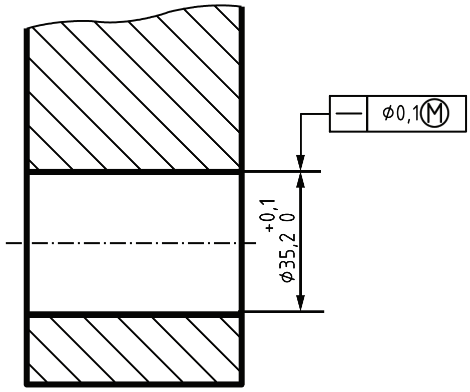
i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature shall have everywhere a local diameter larger than LMS = 34,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 35,0 mm [see rule A 1), 3.4 and 3.5].

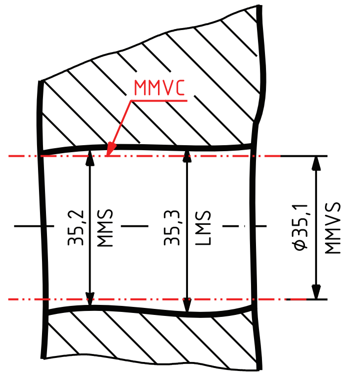
iii) The orientation and location of the MMVC are not controlled by any external constraints (see 3.9, Note 2 to entry).

a) Example of MMR for an external cylindrical feature based on size and form (straightness) requirements

Drawing indication



Interpretation



The intended function of the part toleranced in Figure A.2 b) could be a clearance fit with a shaft of the same length as the toleranced cylindrical feature.

The interpretation is based on the following rules and definitions given in this document.

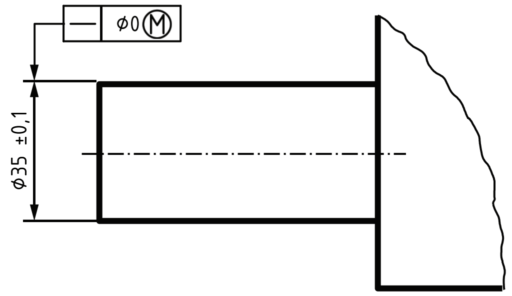
i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature shall have everywhere a local diameter smaller than LMS = 35,3 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 35,2 mm [see rule A 2), 3.4 and 3.5].

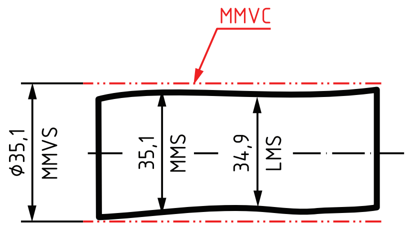
iii) The orientation and location of the MMVC are not controlled by any external constraints (see 3.9, Note 2 to entry).

b) Example of MMR for an internal cylindrical feature based on size and form (straightness) requirements

Drawing indication



Interpretation



The intended function of the part toleranced in Figure A.2 c) could be a clearance fit with a hole of the same length as the toleranced cylindrical feature.

The interpretation is based on the following rules and definitions given in this document.

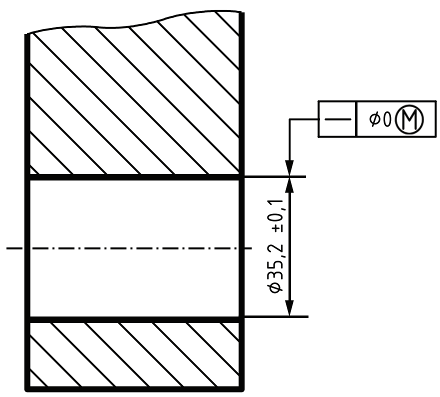
i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature shall have everywhere a local diameter larger than LMS = 34,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 35,1 mm [see rule A 1), 3.4 and 3.5]. The difference between Figures A.2 c) and A.2 a) is in the specification of the local diameter, in this case MMS.

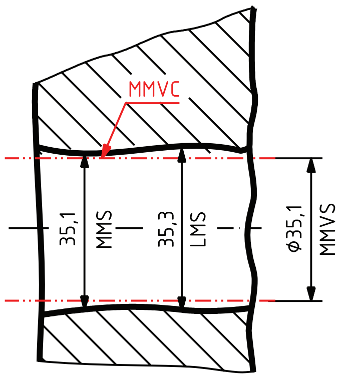
iii) The orientation and location of the MMVC are not controlled by any external constraints (see 3.9, Note 2 to entry).

c) Example of MMR [with 0 Ⓜ] for an external cylindrical feature based on size and form (straightness) requirements

Drawing indication



Interpretation



The intended function of the part toleranced on Figure A.2 d) could be a clearance fit with a shaft of the same length as the toleranced cylindrical feature.

The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

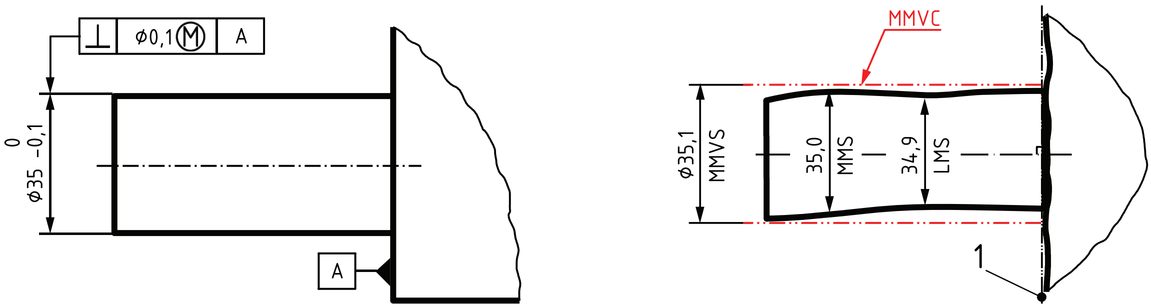
ii) The extracted feature shall have everywhere a local diameter smaller than LMS = 35,3 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 35,1 mm [see rule A 2), 3.4 and 3.5]. The difference between Figures A.2 d) and A.2 b) is in the specification of the local diameter, in this case MMS.

iii) The orientation and location of the MMVC are not controlled by any external constraints (see 3.9, Note 2 to entry).

d) Example of MMR [with 0 Ⓜ] for an internal cylindrical feature based on size and form (straightness) requirements

Figure A.2 — Examples of MMR for a cylindrical feature based on size and form (straightness) requirements

Drawing indication | **Interpretation**



The intended function of the part toleranced in Figure A.3 a) could be an assembly with a part as shown in Figure A.3 b), where the functional requirement is that the two planar faces shall be in contact and the pin shall fit into the hole at the same time.

The interpretation is based on the following rules and definitions given in this document.

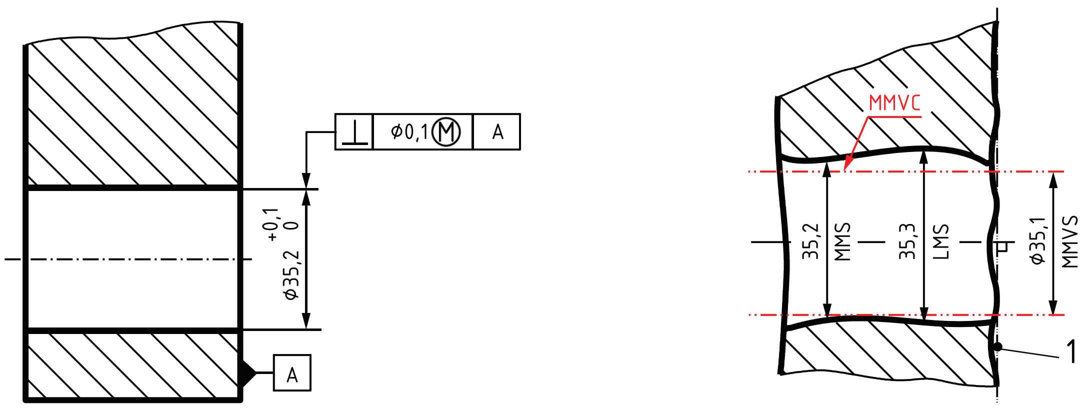
i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature shall have everywhere a local diameter larger than LMS = 34,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 35,0 mm [see rule A 1), 3.4 and 3.5].

iii) The orientation of the MMVC is perpendicular to the datum and the location of the MMVC is not controlled by any external constraints [see rule D and 3.9, Note 2 to entry].

a) Example of MMR for an external cylindrical feature based on size and orientation (perpendicularity) requirements

Drawing indication | **Interpretation**



The intended function of the part toleranced in Figure A.3 b) could be an assembly with a part as shown in Figure A.3 a), where the functional requirement is that the two planar faces shall be in contact and the pin shall fit into the hole at the same time.

The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature shall have everywhere a local diameter smaller than LMS = 35,3 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 35,2 mm [see rule A 2), 3.4 and 3.5].

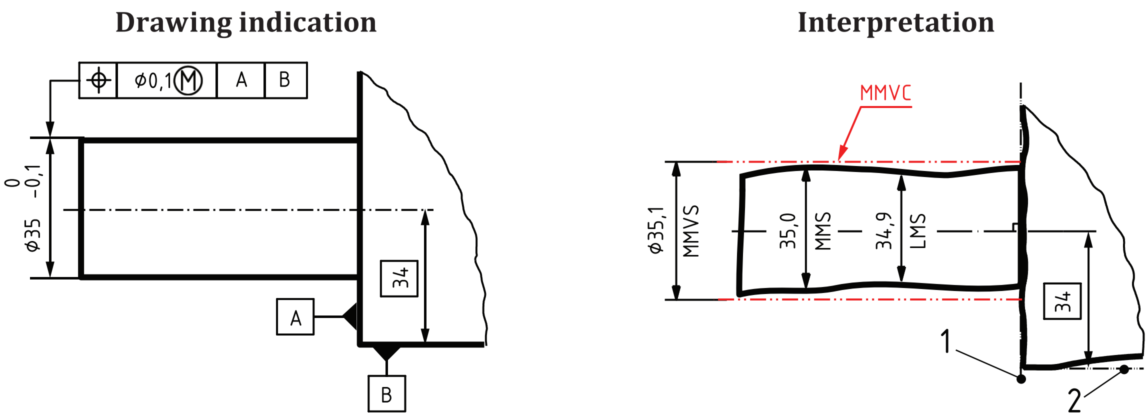
iii) The orientation of the MMVC is perpendicular to the datum and the location of the MMVC is not controlled by any external constraints [see rule D and 3.9, Note 2 to entry].

b) Example of MMR for an internal cylindrical feature based on size and orientation (perpendicularity) requirements

Key

1 datum A

Figure A.3 — Examples of MMR for a cylindrical feature based on size and orientation (perpendicularity) requirements



The intended function of the part toleranced in Figure A.4 a) could be an assembly with a part as shown in Figure A.4 b), where the functional requirement is that the two planar faces A shall be in contact and the two planar faces B shall both simultaneously be in contact with a plane (on a part not shown).

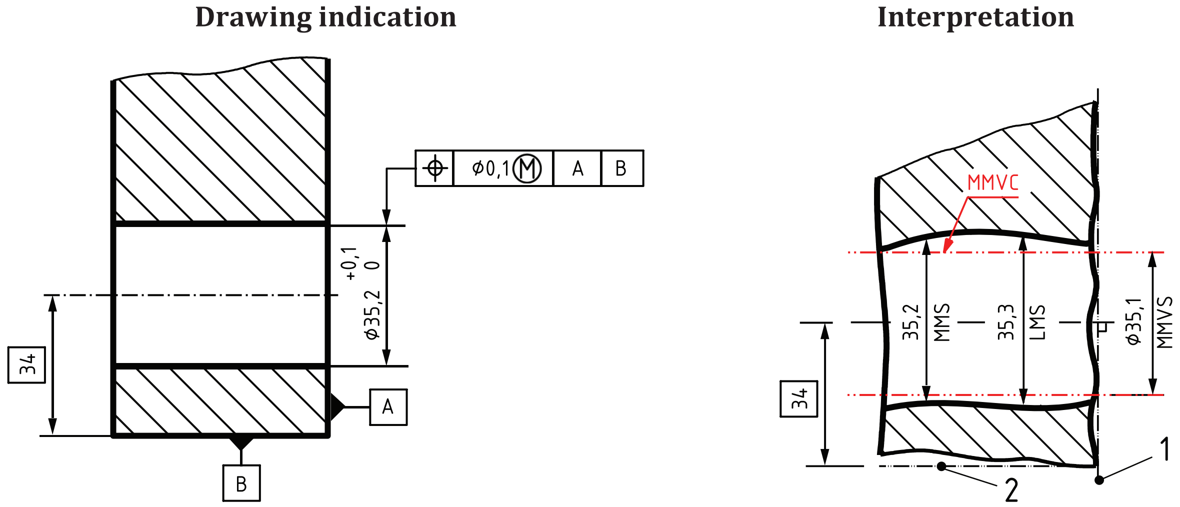
The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature shall have everywhere a local diameter larger than LMS = 34,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 35,0 mm [see rule A 1), 3.4 and 3.5].

iii) The orientation of the MMVC is perpendicular to datum A and the location of the MMVC is in a theoretically exact position 34 mm from datum B [see rule D and 3.9, Note 2 to entry].

a) Example of MMR for an external cylindrical feature based on size and location (position) requirements



The intended function of the part toleranced in Figure A.4 b) could be an assembly with a part as shown in Figure A.4 a), where the functional requirement is that the two planar faces A shall be in contact and the two planar faces B shall both simultaneously be in contact with a plane (on a part not shown).

The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature shall have everywhere a local diameter smaller than LMS = 35,3 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 35,2 mm [see rule A 2), 3.4 and 3.5].

iii) The orientation of the MMVC is perpendicular to datum A and the location of the MMVC is in a theoretically exact position 34 mm from datum B [see rule D and 3.9, Note 2 to entry].

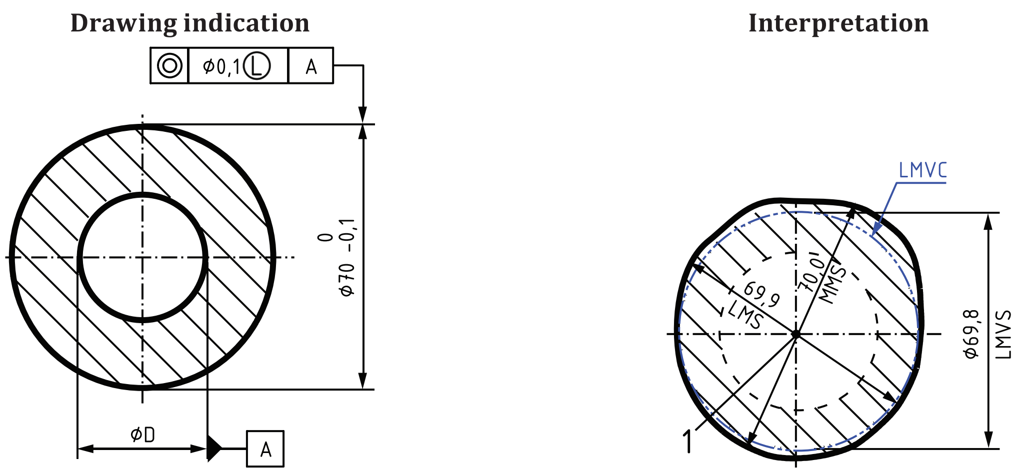
b) Example of MMR for an internal cylindrical feature based on size and location (position) requirements

Key

1 datum A

2 datum B

Figure A.4 — Examples of MMR for a cylindrical feature based on size and location (position) requirements



Position and coaxiality or concentricity can be used with the same meaning in this case.

Figure A.5 a) only illustrates some rules for LMR. This drawing indication is incomplete and does not control the minimum wall thickness. The LMR is missing on the other feature. Therefore, it is not possible to indicate a function.

The interpretation is based on the following rules and definitions given in this document.

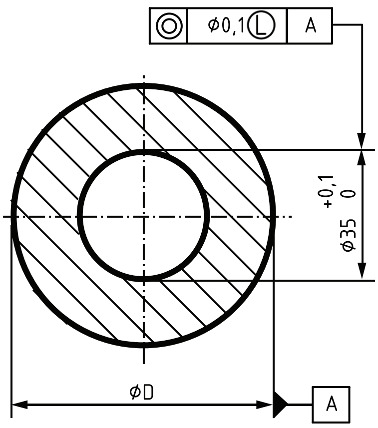
i) The LMVC, which has the diameter LMVS = 69,8 mm, shall be fully contained in the material [see rule J, 3.10 and 3.11].

ii) The extracted feature shall have everywhere a local diameter smaller than MMS = 70,0 mm [see rule I 1), 3.4 and 3.5] and larger than LMS = 69,9 mm [see rule H 1), 3.6 and 3.7].

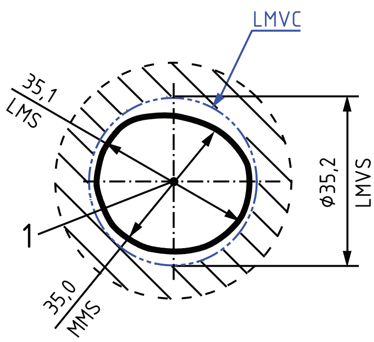
iii) The orientation of the LMVC is parallel to the datum and the location of the LMVC is in a theoretically exact position 0 mm from datum A [see rule K and 3.11, Note 2 to entry].

a) Example 1 of LMR for one external feature of linear size with another concentric internal feature of linear size as the datum

Drawing indication



Interpretation



Position and coaxiality or concentricity can be used with the same meaning in this case.

Figure A.5 b) only illustrates some rules for LMR. This drawing indication is incomplete and does not control the minimum wall thickness. The LMR is missing on the other feature. Therefore, it is not possible to indicate a function.

The interpretation is based on the following rules and definitions given in this document.

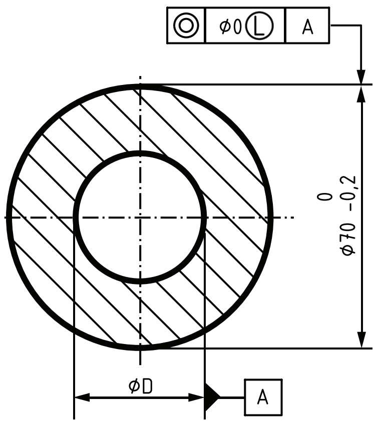
i) The LMVC, which has the diameter LMVS = 35,2 mm, shall be fully contained in the material [see rule J, 3.10 and 3.11].

ii) The extracted feature shall have everywhere a local diameter larger than MMS = 35,0 mm [see rule I 2), 3.4 and 3.5] and smaller than LMS = 35,1 mm [see rule H 2), 3.6 and 3.7].

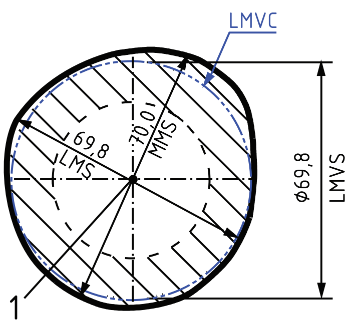
iii) The orientation of the LMVC is parallel to the datum and the location of the LMVC is in a theoretically exact position 0 mm from datum A [see rule K and 3.11, Note 2 to entry].

b) Example 1 of LMR for one internal feature of linear size with another concentric external feature of linear size as the datum

Drawing indication



Interpretation



Position and coaxiality or concentricity can be used with the same meaning in this case.

Figure A.5 c) only illustrates some rules for LMR. This drawing indication is incomplete and does not control the minimum wall thickness. The LMR is missing on the other feature. Therefore, it is not possible to indicate a function.

The interpretation is based on the following rules and definitions given in this document.

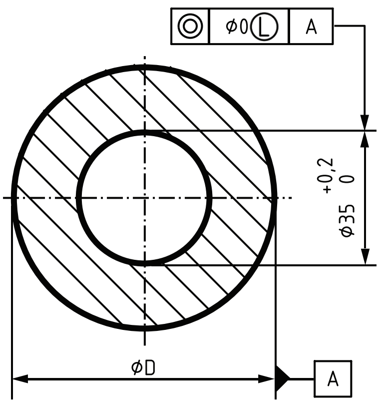
i) The LMVC, which has the diameter LMVS = 69,8 mm, shall be fully contained in the material [see rule J, 3.10 and 3.11].

ii) The extracted feature shall have everywhere a local diameter smaller than MMS = 70,0 mm [see rule I 1), 3.4 and 3.5] and larger than LMS = 69,8 mm [see rule H 1), 3.6 and 3.7]. The difference between Figures A.5 c) and A.5 a) is in the specification of the local diameter, in this case to LMS.

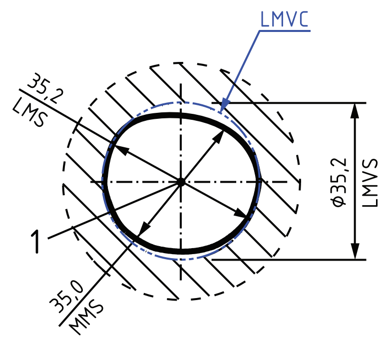
iii) The orientation of the LMVC is parallel to the datum and the location of the LMVC is in a theoretically exact position 0 mm from datum A [see rule K and 3.11, Note 2 to entry].

c) Example 2 of LMR for one external feature of linear size with another concentric internal feature of linear size as the datum

Drawing indication



Interpretation



Position and coaxiality or concentricity can be used with the same meaning in this case.

Figure A.5 d) only illustrates some rules for LMR. This drawing indication is incomplete and does not control the minimum wall thickness. The LMR is missing on the other feature. Therefore, it is not possible to indicate a function.

The interpretation is based on the following rules and definitions given in this document.

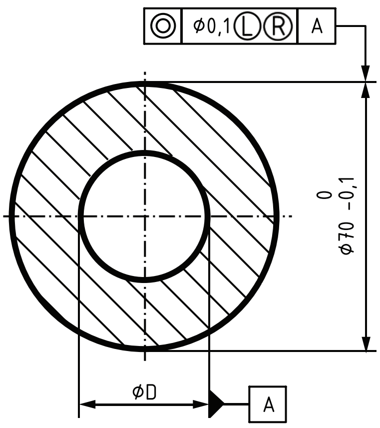
i) The LMVC, which has the diameter LMVS = 35,2 mm, shall be fully contained in the material [see rule J, 3.10 and 3.11].

ii) The extracted feature shall have everywhere a local diameter larger than MMS = 35,0 mm [see rule I 2), 3.4 and 3.5] and smaller than LMS = 35,2 mm [see rule H 2), 3.6 and 3.7]. The difference between Figures A.5 d) and A.5 b) is in the specification of the local diameter, in this case to LMS.

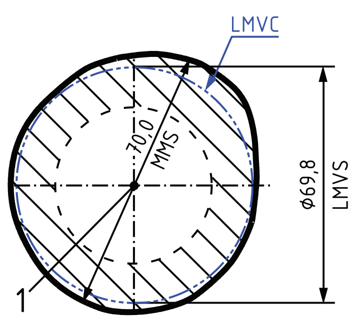
iii) The orientation of the LMVC is parallel to the datum and the location of the LMVC is in a theoretically exact position 0 mm from datum A [see rule K and 3.11, Note 2 to entry].

d) Example 2 of LMR for one internal feature of linear size with another concentric external feature of linear size as the datum

Drawing indication



Interpretation



Position and coaxiality or concentricity can be used with the same meaning in this case.

Figure A.5 e) only illustrates some rules for LMR. This drawing indication is incomplete and does not control the minimum wall thickness. The LMR is missing on the other feature. Therefore, it is not possible to indicate a function.

The interpretation is based on the following rules and definitions given in this document.

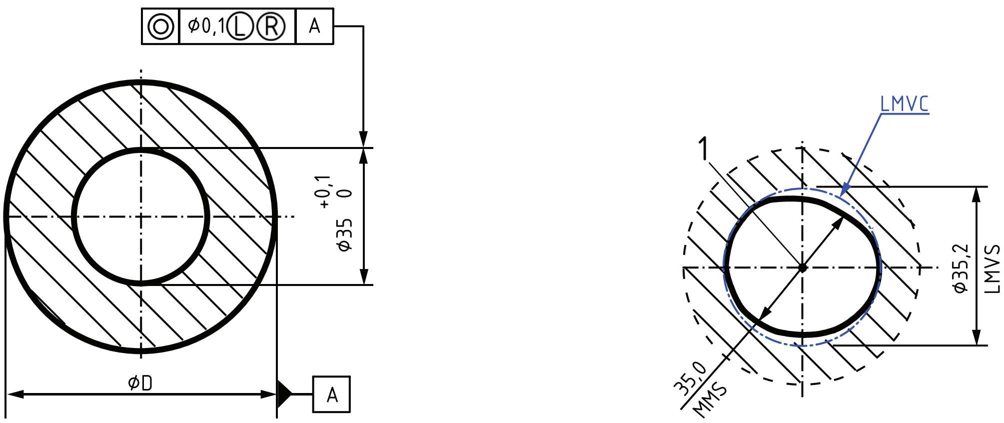
i) The LMVC, which has the diameter LMVS = 69,8 mm, shall be fully contained in the material [see rule J, 3.10 and 3.11].

ii) The extracted feature shall have everywhere a local diameter smaller than MMS = 70,0 mm [see rule I 1), 3.4 and 3.5]. The RPR allows the LLS to decrease below LMS (local diameter may be smaller than 69,9 mm, down to LMVS) [see 5.3].

iii) The orientation of the LMVC is parallel to the datum and the location of the LMVC is in a theoretically exact position 0 mm from datum A [see rule K and 3.11, Note 2 to entry].

e) Example of LMR and RPR for one external feature of linear size with another concentric internal feature of linear size as the datum

Drawing indication **Interpretation**



Position and coaxiality or concentricity can be used with the same meaning in this case.

Figure A.5 f) only illustrates some rules for LMR. This drawing indication is incomplete and does not control the minimum wall thickness. The LMR is missing on the other feature. Therefore, it is not possible to indicate a function.

The interpretation is based on the following rules and definitions given in this document.

i) The LMVC, which has the diameter LMVS = 35,2 mm, shall be fully contained in the material [see rule J, 3.10 and 3.11].

ii) The extracted feature shall have everywhere a local diameter larger than MMS = 35,0 mm [see rule I 2), 3.4 and 3.5]. The RPR allows the ULS to increase beyond LMS (local diameter may be larger than 35,1 mm, up to LMVS) [see 5.3].

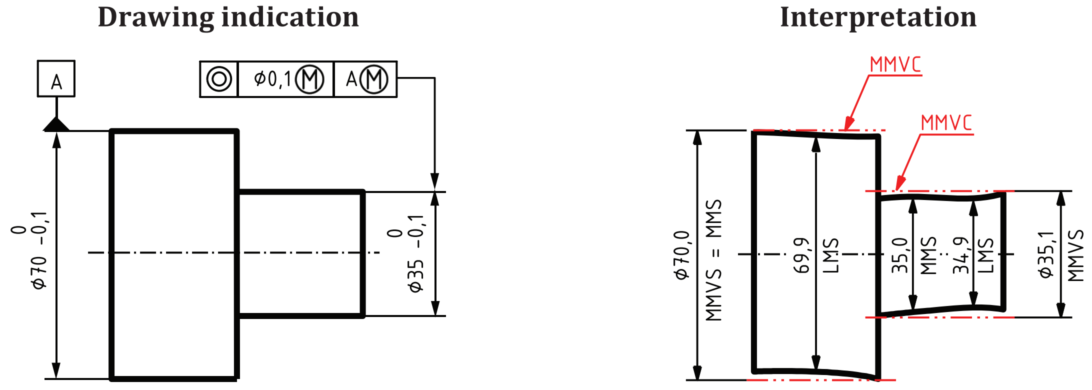
iii) The orientation of the LMVC is parallel to the datum and the location of the LMVC is in a theoretically exact position 0 mm from datum A [see rule K and 3.11, Note 2 to entry].

f) Example of LMR and RPR for one internal feature of linear size with another concentric external feature of linear size as the datum

Key

1 datum A

Figure A.5 — Examples of LMR for one feature of linear size with another concentric feature of linear size as the datum



The intended function of the part toleranced in Figure A.6 a) could be an assembly with a part similar to that of Figure A.6 b).

The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

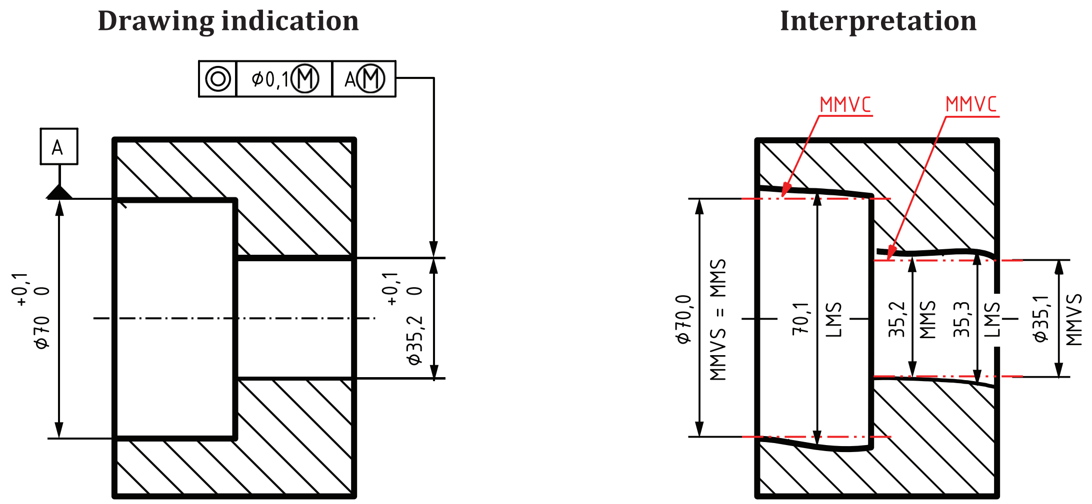
ii) The extracted feature of the toleranced feature shall have everywhere a local diameter larger than LMS = 34,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 35,0 mm [see rule A 1), 3.4 and 3.5].

iii) The location of the MMVC is at 0 mm from the axis of the MMVC of the datum feature [see rule D and 3.9, Note 2 to entry].

iv) The extracted feature of the datum feature shall not violate the MMVC, which has the diameter MMVS = MMS = 70,0 mm [see rules E and F, 3.8 and 3.9].

v) The extracted feature of the datum feature shall have everywhere a local diameter larger than LMS = 69,9 mm [see rule B 1), 3.6 and 3.7].

a) Example of MMR for an external cylindrical feature based on size and location (coaxiality) requirements with the axis of a cylindrical feature – with a size requirement – as the datum and also with MMR



The intended function of the part toleranced in Figure A.6 b) could be an assembly with a part similar to that of Figure A.6 a).

The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

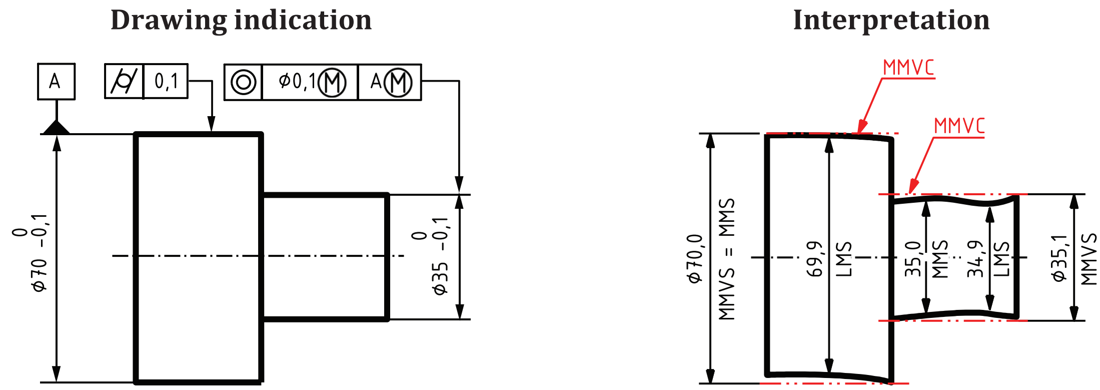
ii) The extracted feature of the toleranced feature shall have everywhere a local diameter smaller than LMS = 35,3 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 35,2 mm [see rule A 2), 3.4 and 3.5].

iii) The location of the MMVC is at 0 mm from the axis of the MMVC of the datum feature [see rule D and 3.9, Note 2 to entry].

iv) The extracted feature of the datum feature shall not violate the MMVC, which has the diameter MMVS = MMS = 70,0 mm [see rules E and F, 3.10 and 3.11].

v) The extracted feature of the datum feature shall have everywhere a local diameter smaller than LMS = 70,1 mm [see rule B 2), 3.6 and 3.7].

b) Example of MMR for an internal cylindrical feature based on size and location (coaxiality) requirements with the axis of a cylindrical feature – with a size requirement – as the datum and also with MMR



The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature of the toleranced feature shall have everywhere a local diameter larger than LMS = 34,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 35,0 mm [see rule A 1), 3.4 and 3.5].

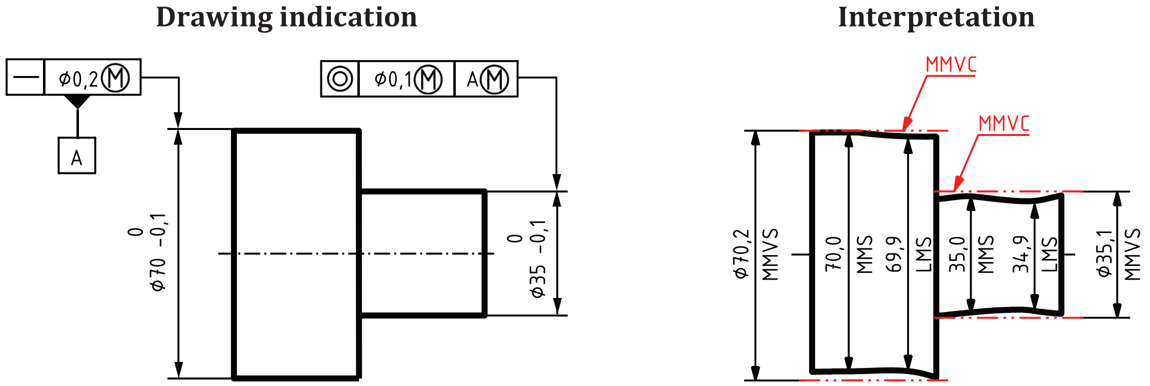
iii) The location of the MMVC is at 0 mm from the axis of the MMVC of the datum feature [see rule D and 3.9, Note 2 to entry].

iv) The extracted feature of the datum feature shall not violate the MMVC, which has the diameter MMVS = MMS = 70,0 mm [see rules E and F, 3.8 and 3.9]. The cylindricity specification has no effect on datum MMVS as specified by rule F.

v) The extracted feature of the datum feature shall have everywhere a local diameter larger than LMS = 69,9 mm [see rule B 1), 3.6 and 3.7].

c) Example of MMR for an external cylindrical feature based on size and location (coaxiality) requirements with the axis of a cylindrical feature – with a size requirement – as the datum with geometrical tolerance and also with MMR

Figure A.6 — Examples of MMR for a cylindrical feature based on size and location (coaxiality) requirements with the axis of a cylindrical feature – with a size requirement – as the datum and also with MMR



The intended function of the part toleranced in Figure A.7 a) could be an assembly with a part similar to that of Figure A.7 b).

The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

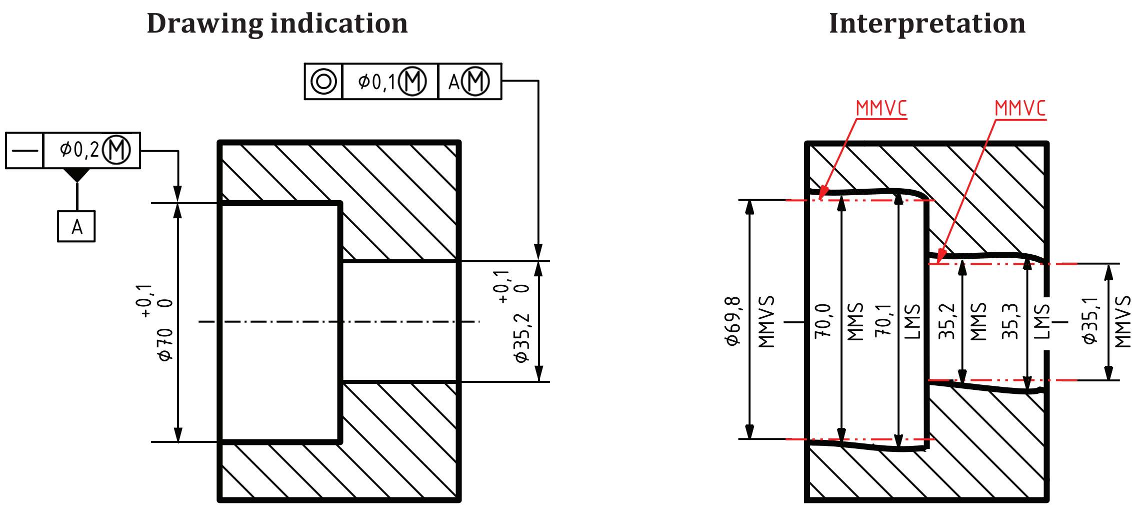
ii) The extracted feature of the toleranced feature shall have everywhere a local diameter larger than LMS = 34,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 35,0 mm [see rule A 1), 3.4 and 3.5].

iii) The location of the MMVC is at 0 mm from the axis of the MMVC of the datum feature [see rule D and 3.9, Note 2 to entry].

iv) The extracted feature of the datum feature shall not violate the MMVC, which has the diameter MMVS = 70,2 mm [see rules E and G, 3.8 and 3.9]. See also B.3 for former practice.

v) The extracted feature of the datum feature shall have everywhere a local diameter larger than LMS = 69,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 70,0 mm [see rule A 1), 3.4 and 3.5].

a) Example of MMR for an external cylindrical feature based on size and location (coaxiality) requirements with the axis of a cylindrical feature – with a size requirement and a form tolerance – as the datum and also with MMR



The intended function of the part toleranced in Figure A.7 b) could be an assembly with a part with similar shape to the one in Figure A.7 a) but with appropriate dimensions.

The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature of the toleranced feature shall have everywhere a local diameter smaller than LMS = 35,3 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 35,2 mm [see rule A 2), 3.4 and 3.5].

iii) The location of the MMVC is at 0 mm from the axis of the MMVC of the datum feature [see rule D and 3.9, Note 2 to entry].

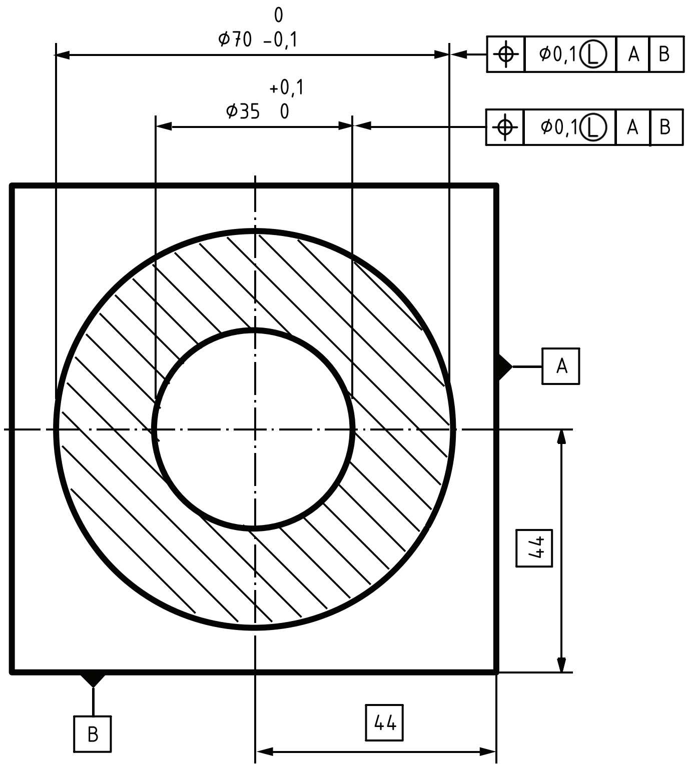
iv) The extracted feature of the datum feature shall not violate the MMVC, which has the diameter MMVS = 69,8 mm [see rules E and G, 3.8 and 3.9].

v) The extracted feature of the datum feature shall have everywhere a local diameter smaller than LMS = 70,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 70 mm [see rule A 2), 3.4 and 3.5].

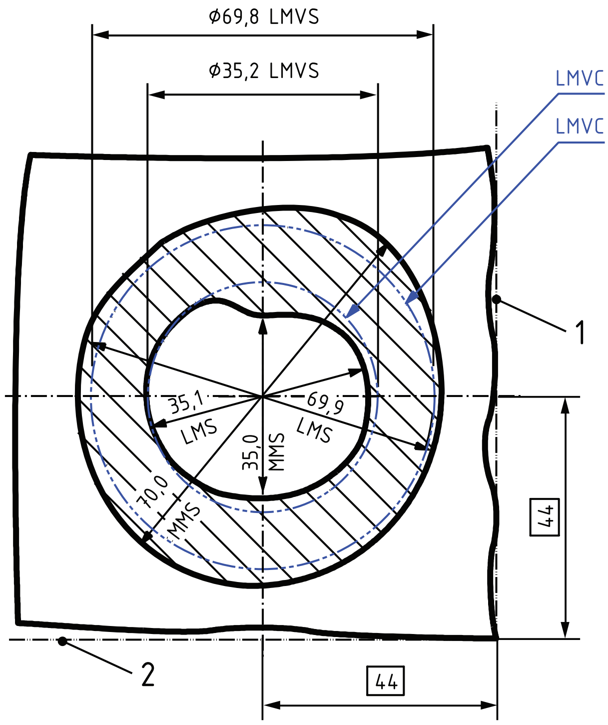
b) Example of MMR for an internal cylindrical feature based on size and location (coaxiality) requirements with the axis of a cylindrical feature – with a size requirement and a form tolerance – as the datum and also with MMR

Figure A.7 — Examples of MMR for a cylindrical feature based on size and location (coaxiality) requirements with the axis of a cylindrical feature – with a size requirement and a form tolerance – as the datum and also with MMR

Drawing indication



Interpretation



The interpretation is based on the following rules and definitions given in this document.

i) The LMVC of the external feature shall be fully contained in the material. The diameter LMVS = 69,8 mm [see rule J, 3.10 and 3.11].

ii) The extracted feature of the external feature shall have everywhere a local size smaller than MMS = 70,0 mm (see rule I 1), 3.4 and 3.5) and larger than LMS = 69,9 mm [see rule H 1), 3.6 and 3.7].

iii) The LMVC of the internal feature shall be fully contained in the material. The diameter LMVS = 35,2 mm [see rule J, 3.10 and 3.11].

iv) The extracted feature of the internal feature shall have everywhere a local size larger than MMS = 35,0 mm (see rule I 2), 3.4 and 3.5) and smaller than LMS = 35,1 mm [see rule H 2), 3.6 and 3.7].

v) The LMVC of both the external and the internal feature shall be in a theoretically exact orientation relative to the datum system and the location relative to the datum system shall be at a position 44 × 44 mm [see rule K and 3.11, Note 2 to entry].

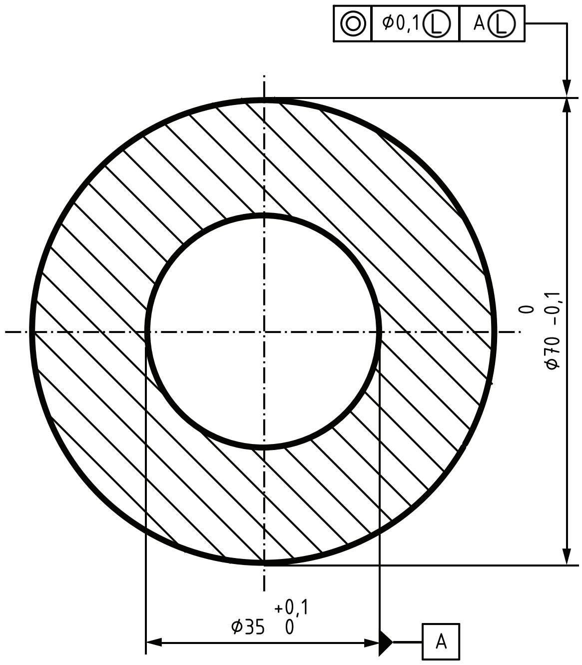
Key

1 datum A

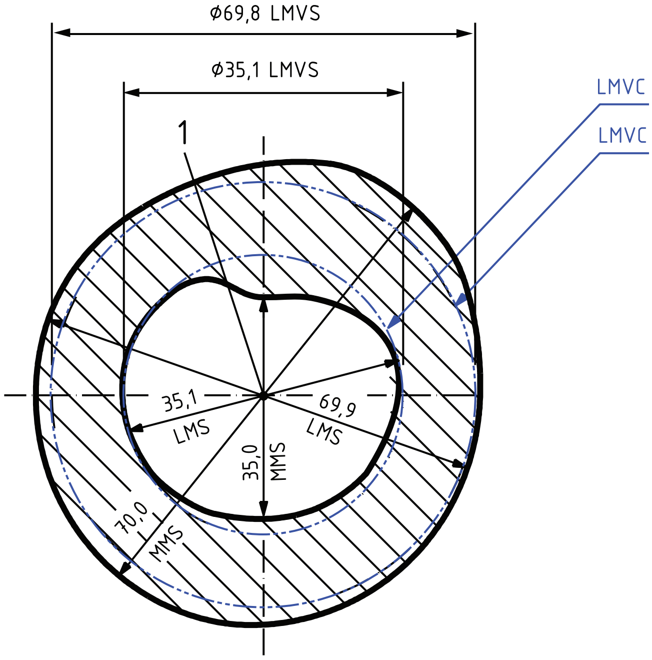
2 datum B

Figure A.8 — Example of LMR for two concentric cylindrical features (internal and external) both controlled by size and location (position) to the same datum system A and B

Drawing indication



Interpretation



The interpretation is based on the following rules and definitions given in this document.

i) The LMVC of the external feature shall be fully contained in the material. The diameter LMVS = 69,8 mm [see rule J, 3.10 and 3.11].

ii) The extracted feature of the external feature shall have everywhere a local size smaller than MMS = 70,0 mm [see rule I 1), 3.4 and 3.5] and larger than LMS = 69,9 mm [see rule H 1), 3.6 and 3.7].

iii) The LMVC of the internal datum feature shall be fully contained in the material. The diameter LMVS = LMS = 35,1 mm [see rule L and M, 3.10 and 3.11].

iv) The extracted feature of the internal feature shall have everywhere a local size larger than MMS = 35,0 mm [see rule I 2), 3.4 and 3.5] and smaller than LMS = 35,1 mm [see rule H 2), 3.6 and 3.7].

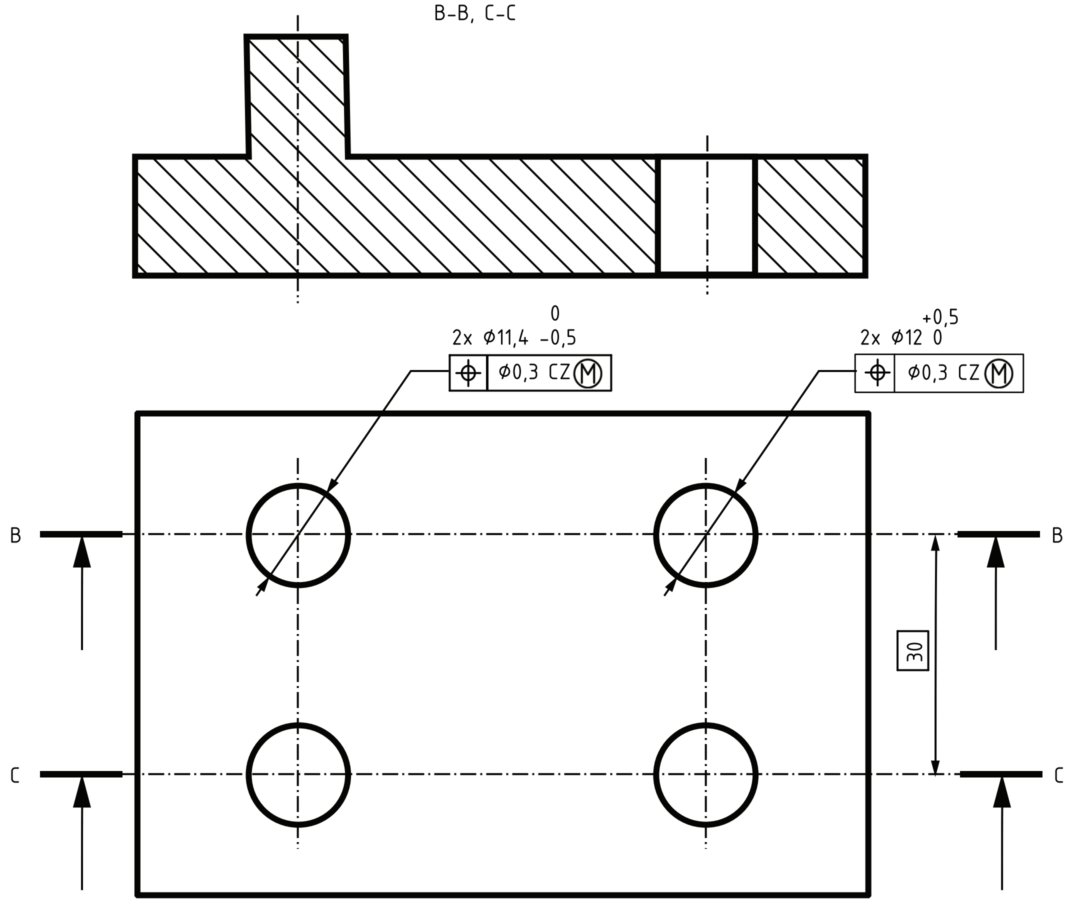
v) The LMVC of the external feature shall be in a theoretically exact location at 0 mm from the axis of the LMVC of the internal datum feature [see rule K and 3.11, Note 2 to entry].

Key

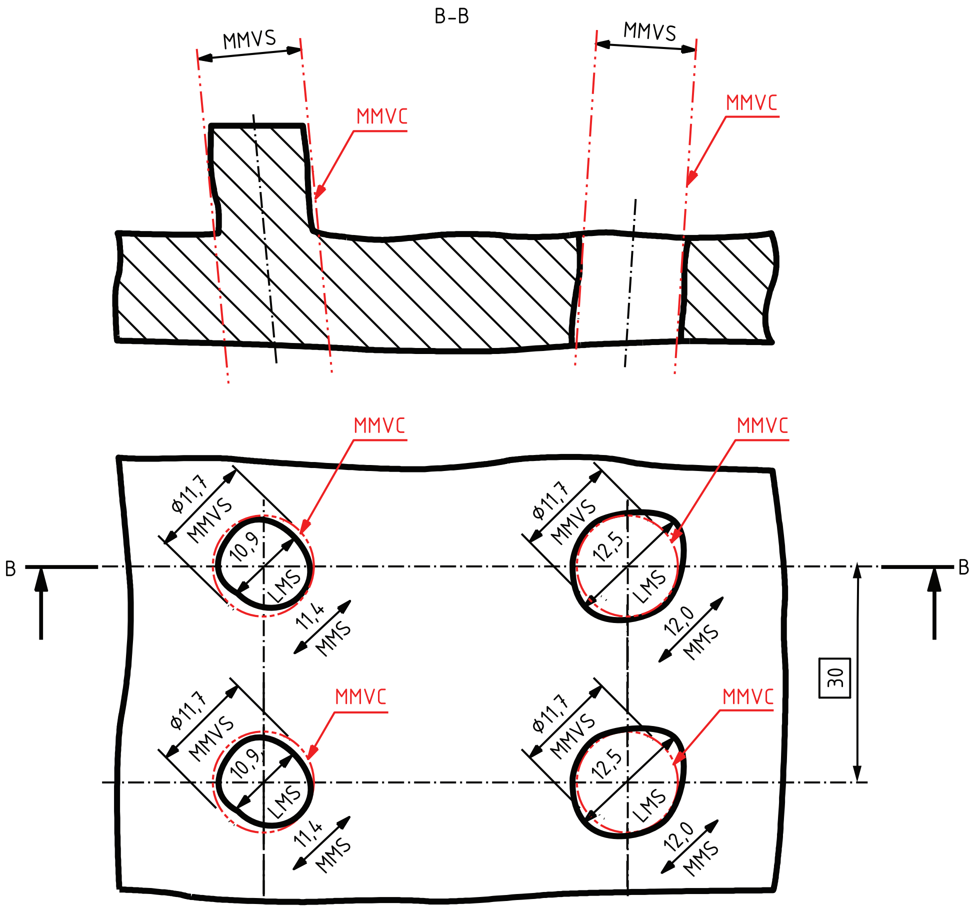
1 datum A

Figure A.9 — Example of LMR for an external cylindrical feature controlled by size and location (coaxiality) relative to an internal cylindrical feature used as a datum controlled by size and LMR

Drawing indication



Interpretation



The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced pins shall not violate the MMVC, which has the diameter MMVS = 11,7 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature of the toleranced pins shall have everywhere a local diameter larger than LMS = 10,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 11,4 mm [see rule A 1), 3.4 and 3.5].

iii) The extracted feature of the toleranced holes shall not violate the MMVC, which has the diameter MMVS = 11,7 mm [see rule C, 3.9 and 4.1.2].

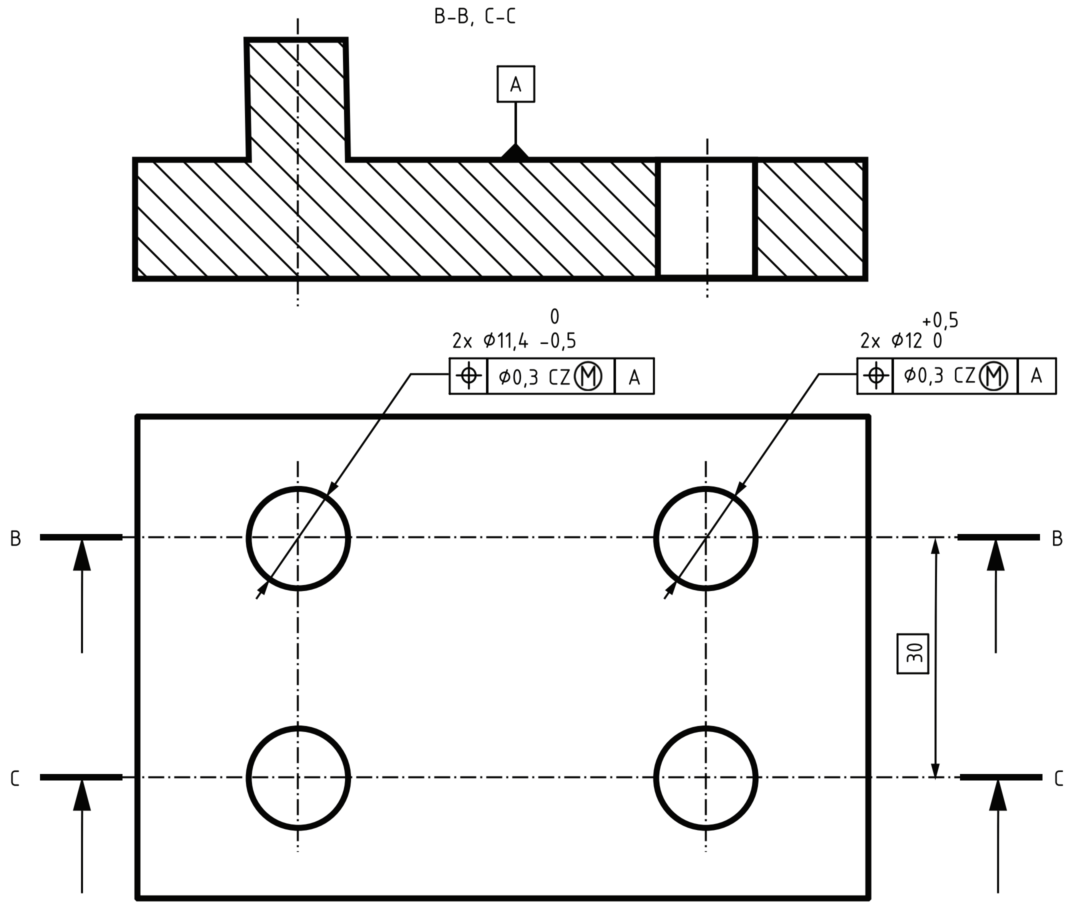
iv) The extracted feature of the toleranced holes shall have everywhere a local diameter smaller than LMS = 12,5 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 12,0 mm [see rule A 2), 3.4 and 3.5].

v) The two MMVCs of the toleranced pins are in theoretically exact orientation and location: parallel to each other and at a distance 30 mm relative to each other [see rule D and 3.9, Note 2 to entry]. There is no orientation or location requirement relative to the rest of the part.

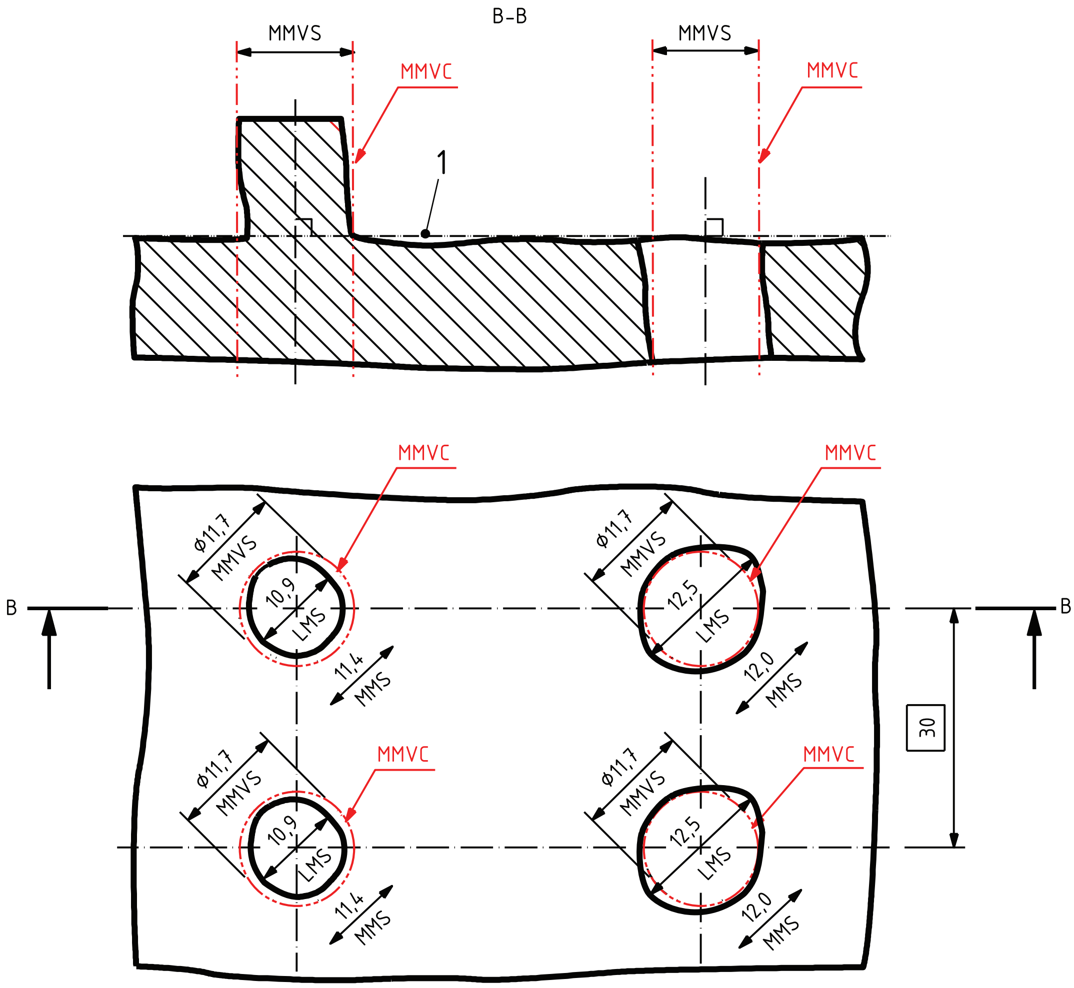
vi) The two MMVCs of the toleranced holes are in theoretically exact orientation and location: parallel to each other and at a distance 30 mm relative to each other [see rule D and 3.9, Note 2 to entry]. There is no orientation or location requirement relative to the rest of the part, and more particularly between the holes and the pins.

Figure A.10 — Example of MMR separately applied on a pattern of holes and on a pattern of pins, without using a datum

Drawing indication



Interpretation



The interpretation is based on the following rules and definitions given in this document.

i)	The extracted feature of the toleranced pins shall not violate the MMVC, which has the diameter MMVS = 11,7 mm [see rule C, 3.9 and 4.1.2].
ii)	The extracted feature of the toleranced pins shall have everywhere a local diameter larger than LMS = 10,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 11,4 mm [see rule A 1), 3.4 and 3.5].
iii)	The extracted feature of the toleranced holes shall not violate the MMVC, which has the diameter MMVS = 11,7 mm [see rule C, 3.9 and 4.1.2].
iv)	The extracted feature of the toleranced holes shall have everywhere a local diameter smaller than LMS = 12,5 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 12,0 mm [see rule A 2), 3.4 and 3.5].
v)	The two MMVCs of the toleranced pins are in theoretically exact orientation and location: perpendicular to the datum A and at a distance 30 mm relative to each other [see rule D and 3.9, Note 2 to entry].

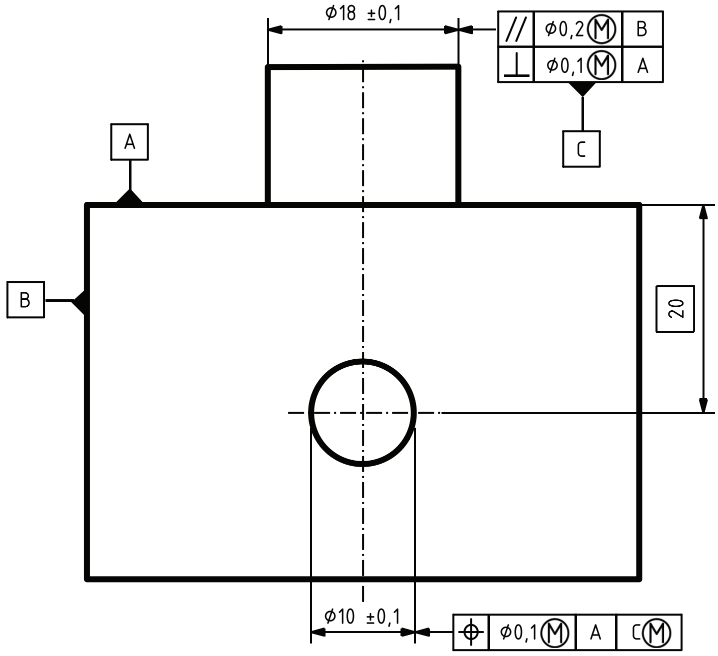
vi) The two MMVCs of the toleranced holes are in theoretically exact orientation and location: perpendicular to the datum A and at a distance 30 mm relative to each other [see rule D and 3.9, Note 2 to entry]. There is no orientation or location requirement relative to the pins.

Key

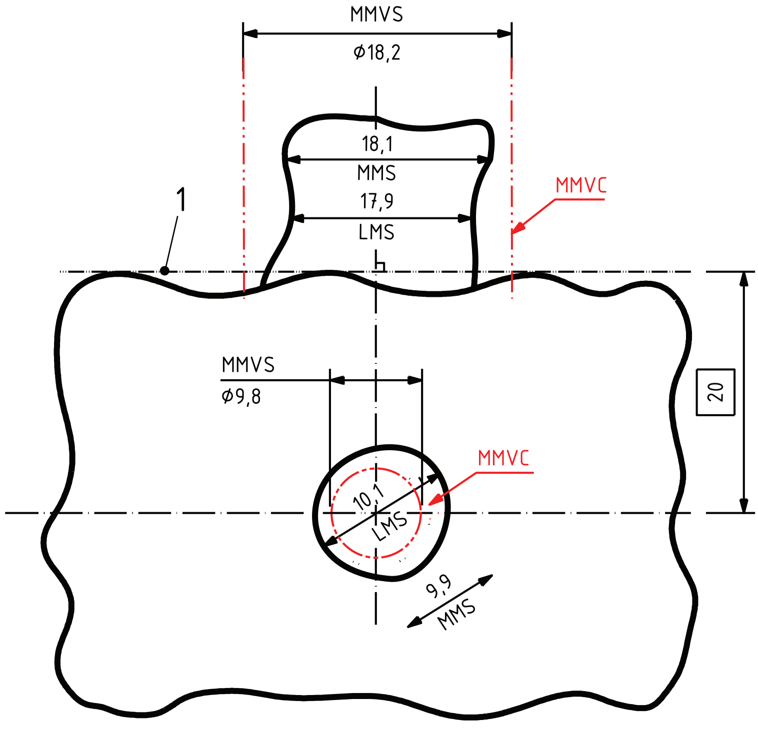
1 datum A

Figure A.11 — Example of MMR separately applied on a pattern of holes and on a pattern of pins with using a datum

Drawing indication



Interpretation



The interpretation is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced hole shall not violate the MMVC, which has the diameter MMVS = 9,8 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature of the toleranced hole shall have everywhere a local diameter smaller than LMS = 10,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 9,9 mm [see rule A 2), 3.4 and 3.5].

iii) The location of the MMVC of the hole is at 20 mm from the datum plane A and at 0 mm from the axis of the MMVC of the datum feature C [see rule D and 3.9, Note 2 to entry].

iv) The extracted feature of the datum feature C shall not violate the MMVC, which has the diameter MMVS = 18,2 mm [see rules E and G, 3.8 and 3.9].

v) The extracted feature of the datum feature C shall have everywhere a local diameter larger than LMS = 17,9 mm [see rule B 1), 3.6 and 3.7] and smaller than MMS = 18,1 mm [see rule A 1), 3.4 and 3.5].

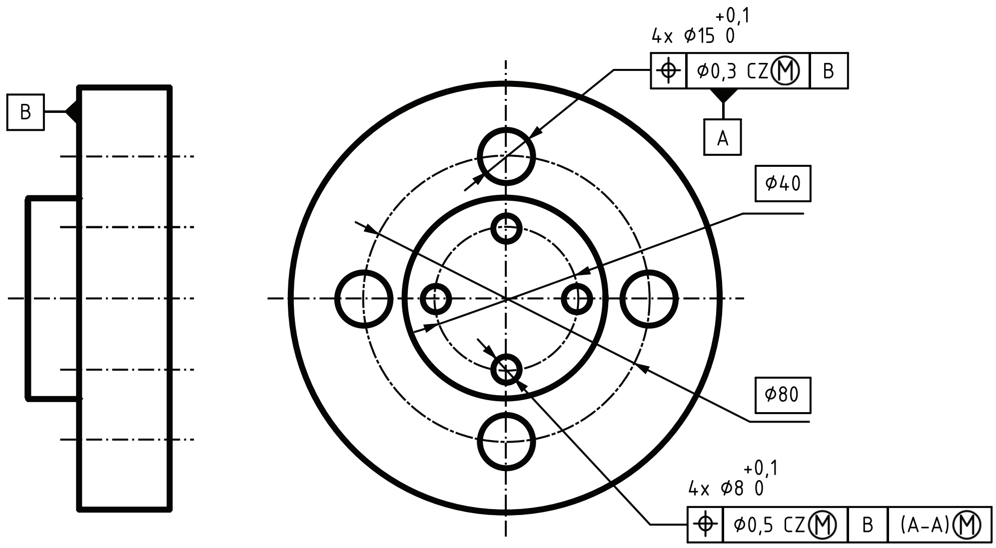
vi) The MMVC of the datum feature C is in theoretically exact orientation relative to the datum A, i.e. perpendicular to the datum plane A [see rule D and 3.9, Note 2 to entry]. There is no other orientation or location requirement for the datum feature C. Parallelism relative to datum plane B is another geometrical specification that shall be respected according to the independency principle, but does not concern the pin as datum feature C.

Key

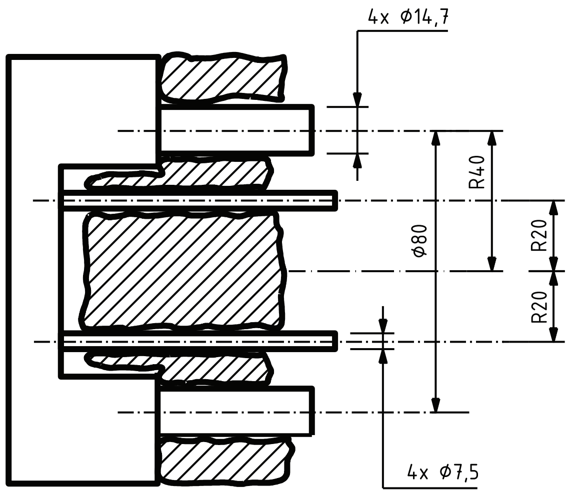
1 datum A

Figure A.12 — Example of use of MMR applied on a datum feature (followed by the symbol Ⓜ) controlled by an orientation specification whose tolerance value is followed by the symbol Ⓜ and whose datum corresponds exactly to the datum called before the related datum

Drawing indication



Interpretation



The interpretation is based on the following rules and definitions given in this document.

i) The extracted features of the four toleranced features (diameter 8 mm) shall not violate the MMVCs, which have the diameters MMVS = 7,5 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted features of the four toleranced features (diameter 8 mm) shall have everywhere a local diameter smaller than LMS = 8,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 8 mm [see rule A 2), 3.4 and 3.5].

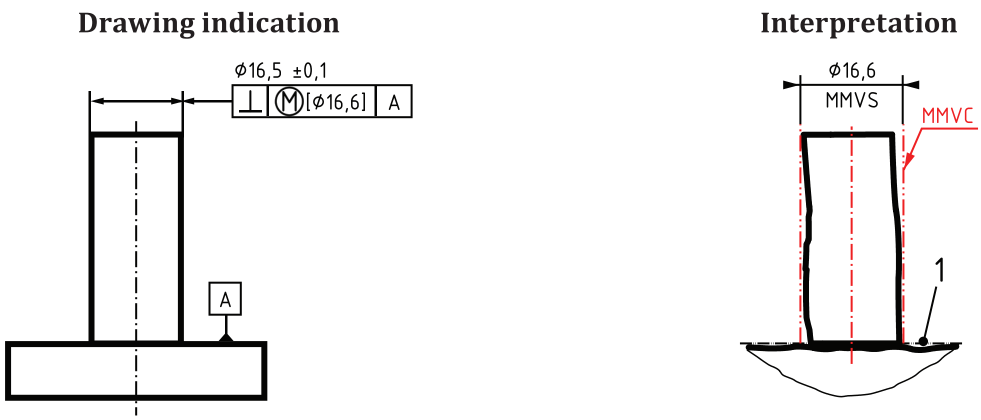
iii) The locations of the four MMVCs are positioned (and oriented) to the situation features of the collection of associated features (constrained at a fixed size equal to their MMVS) to the datum features [see rule D and 3.9, Note 2 to entry]. The location and orientation of the MMVCs are theoretically exact.

iv) The extracted features of the datum features (diameter 15 mm) shall not violate their MMVCs, which are four cylinders with diameter MMVS = 15,0 mm − 0,3 mm = 14,7 mm [see rules E and G, 3.8 and 3.9].

v) The extracted features of the datum features (diameter 15 mm) shall have everywhere a local diameter smaller than LMS = 15,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 15,0 mm.

vi) MMVCs of the four datum features are in theoretically exact orientation relative to datum B, i.e. perpendicular to the datum plane B, and in theoretically exact location relative to each other, i.e. equally distributed on a cylinder of diameter 80 mm [see rule D, 3.9, Note 2 to entry and 4.2.2].

Figure A.13 — Example of use of MMR applied on a common datum



The interpretation of the tolerance indicator is based on the following rules and definitions given in this document.

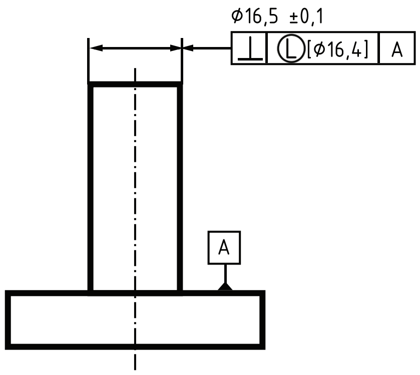
i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 16,6 mm [see rule C, 3.9 and 4.2.3].

ii) The orientation of the MMVC is perpendicular to the datum and the location of the MMVC is not constrained [see rule D and 3.9, Note 2 to entry].

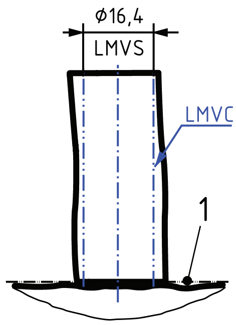
The size specification is not part of the MMR. However, the size specification has to be taken into account as a separate independent specification according to ISO 14405-1 (see 4.1.3).

a) Direct indication of MMVS on an external feature

Drawing indication



Interpretation



The interpretation of the tolerance indicator is based on the following rules and definitions given in this document.

i) The extracted feature shall not violate the LMVC, which has the diameter LMVS = 16,4 mm [see rule J, 3.10, 3.11 and 4.3.3].

ii) The orientation of the LMVC is perpendicular to the datum and the location of the LMVC is not constrained [see rule K and 3.11, Note 2 to entry].

The size specification is not part of the LMR. However, the size specification has to be taken into account as a separate independent specification according to ISO 14405-1 (see 4.1.3).

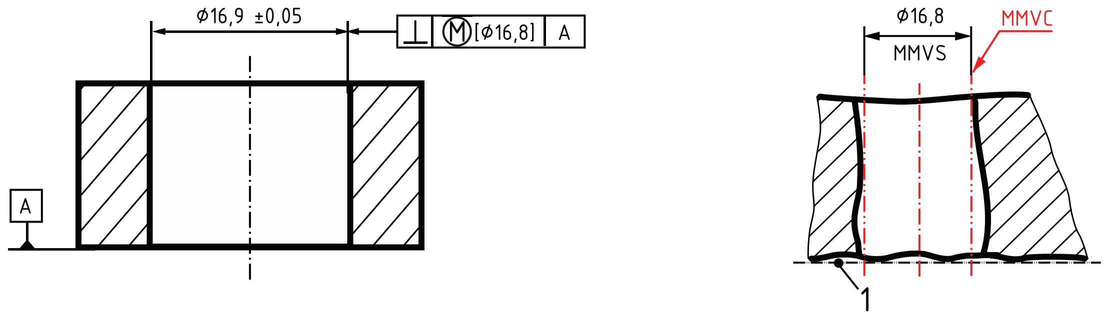
b) Direct indication of LMVS on an external feature

Key

1 datum A

Figure A.14 — Direct indication of MMVS and LMVS on an external feature

Drawing indication **Interpretation**



The interpretation of the tolerance indicator is based on the following rules and definitions given in this document.

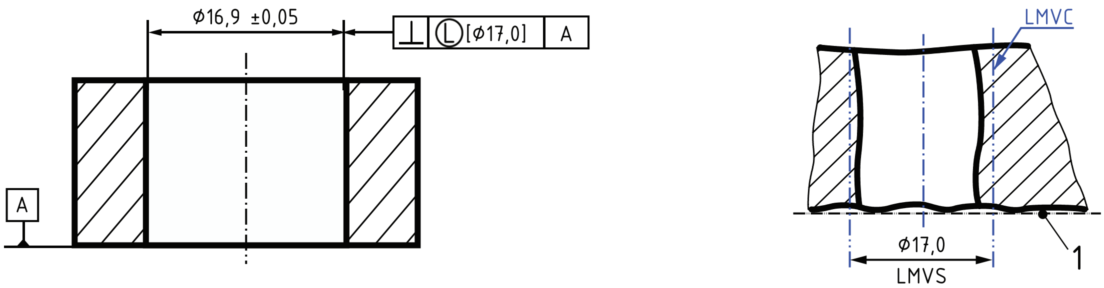
i) The extracted feature shall not violate the MMVC, which has the diameter MMVS = 16,8 mm [see rule C, 3.9 and 4.2.3].

ii) The orientation of the MMVC is perpendicular to the datum and the location of the MMVC is not constrained [see rule D and 3.9, Note 2 to entry].

The size specification is not part of the MMR. However, the size specification has to be taken into account as a separate independent specification according to ISO 14405-1 (see 4.1.3).

a) Direct indication of MMVS on an internal feature

Drawing indication **Interpretation**



The interpretation of the tolerance indicator is based on the following rules and definitions given in this document.

i) The extracted feature shall not violate the LMVC, which has the diameter LMVS = 17,0 mm [see rule J, 3.10, 3.11 and 4.3.3].

ii) The orientation of the LMVC is perpendicular to the datum and the location of the LMVC is not constrained [see rule K and 3.11, Note 2 to entry].

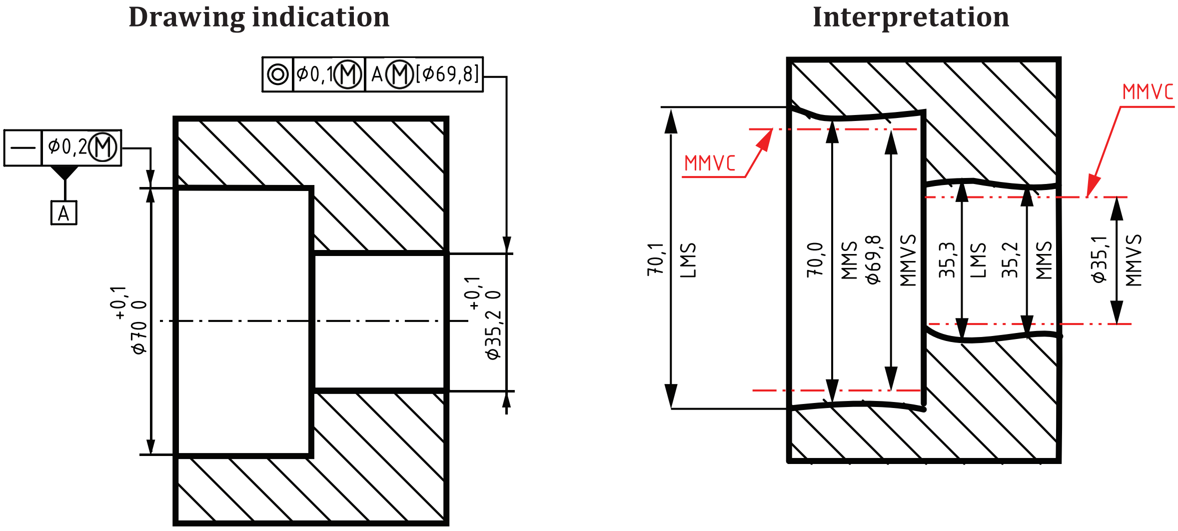
The size specification is not part of the LMR. However, the size specification has to be taken into account as a separate independent specification according to ISO 14405-1 (see 4.1.3).

b) Direct indication of LMVS on an internal feature

Key

1 datum A

Figure A.15 — Direct indication of MMVS or LMVS on an internal feature



The interpretation of the tolerance indicator with coaxiality is based on the following rules and definitions given in this document.

i) The extracted feature of the toleranced feature shall not violate the MMVC, which has the diameter MMVS = 35,1 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted feature of the toleranced feature shall have everywhere a local diameter smaller than LMS = 35,3 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 35,2 mm [see rule A 2), 3.4 and 3.5].

iii) The location of the MMVC is at 0 mm from the axis of the MMVC of the datum feature [see rule D and 3.9, Note 2 to entry].

iv) The extracted feature of the datum feature shall not violate the MMVC, which has the diameter MMVS = 69,8 mm [see rule E, 3.8, 3.9 and rule O].

The interpretation of the tolerance indicator with straightness is based on the following rules and definitions given in this document.

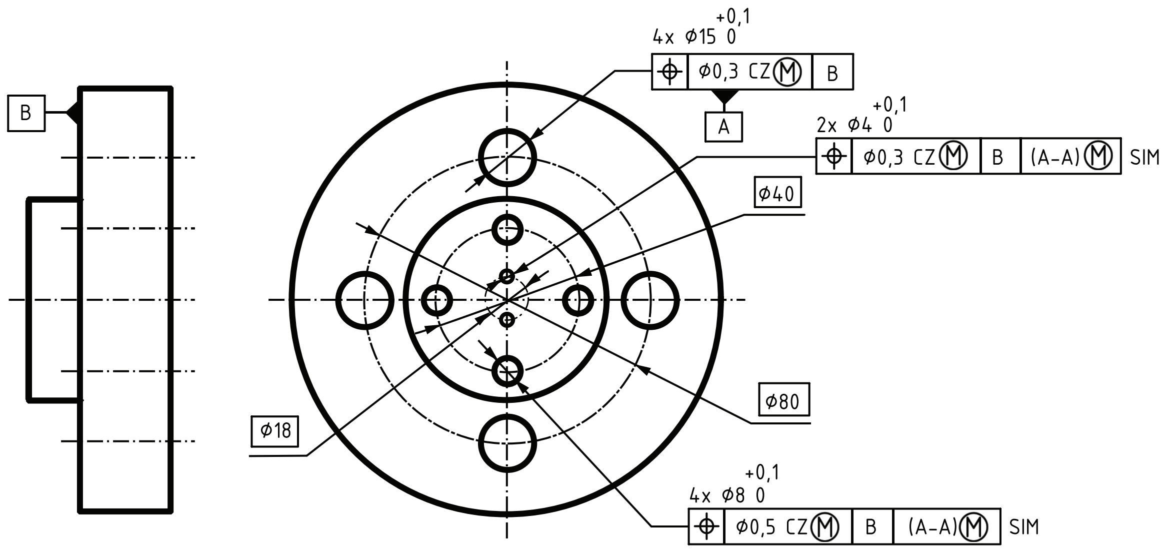
a) The extracted feature shall not violate the MMVC, which has a diameter MMVS = 70 − 0,2 = 69,8 mm [see rule C, 3.9 and 4.1.2].

b) The extracted feature shall have everywhere a local diameter smaller than LMS = 70,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 70,0 mm [see rule A 2), 3.4 and 3.5].

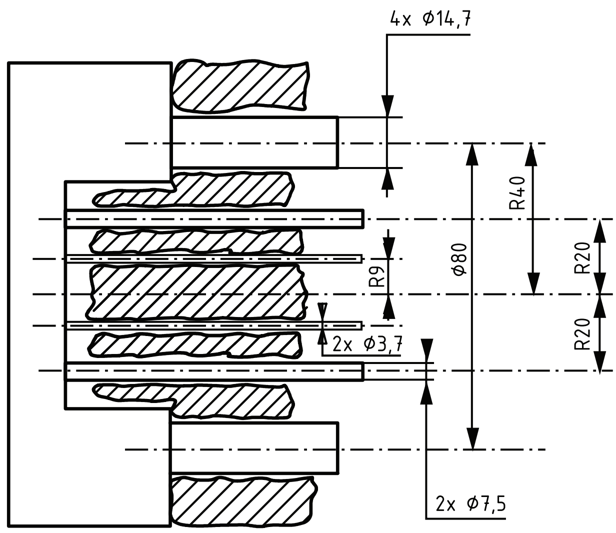
c) The orientation and location of the MMVC are not constrained [see 3.9, Note 2 to entry].

Figure A.16 — Direct indication of MMVS on an internal feature datum

Drawing indication



Interpretation



The interpretation is based on the following rules and definitions given in this document.

i) The extracted features of the four toleranced features (diameter 8 mm) shall not violate the MMVCs, which have the diameters MMVS = 7,5 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted features of the four toleranced features (diameter 8 mm) shall have everywhere a local diameter smaller than LMS = 8,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 8,0 mm [see rule A 2), 3.4 and 3.5].

iii) The four MMVCs are positioned (and oriented) to the situation features of the collection of associated features (constrained at a fixed size equal to their MMVS) to the datum features [see rule D and 3.9, Note 2 to entry]. The location and orientation of the MMVCs are theoretically exact.

iv) The extracted features of the datum features (diameter 15 mm) shall not violate their MMVCs, which are four cylinders with diameter MMVS = 15,0 − 0,3 = 14,7 mm [see rules E and G, 3.8 and 3.9].

v) The extracted features of the datum features (diameter 15 mm) shall have everywhere a local diameter smaller than LMS = 15,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 15,0 mm.

vi) MMVCs of the four datum features (diameter 15 mm) are in theoretically exact orientation relative to datum B, i.e. perpendicular to the datum plane B, and in theoretically exact location relative to each other, i.e. equally distributed on a cylinder of diameter 80 mm [see rule D, 3.9, Note 2 to entry and 4.2.2].

vii) The extracted features of the two toleranced features (diameter 4 mm) shall not violate the MMVCs, which have the diameters MMVS = 3,7 mm [see rule C, 3.9 and 4.1.2].

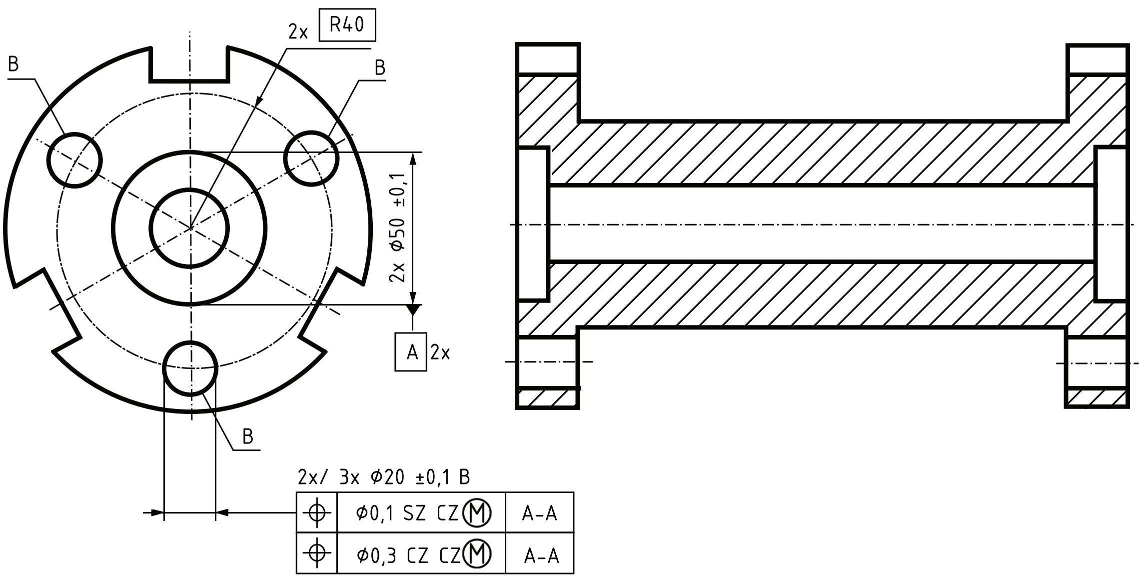
viii) The extracted features of the two toleranced features (diameter 4 mm) shall have everywhere a local diameter smaller than LMS = 4,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 4,0 mm [see rule A 2), 3.4 and 3.5].

ix) The locations of the two MMVCs are positioned (and oriented) to the situation features of the collection of associated features (constrained at a fixed size equal to their MMVS) to the datum features [see rule D and 3.9, Note 2 to entry]. The location and orientation of the MMVCs are theoretically exact.

x) The specified datums with two simultaneous geometrical specifications (with SIM symbol) are identical. The MMVC corresponding to the datum of diameter 8 mm hole geometrical specification has the same orientation and location as the MMVC corresponding to the datum of diameter 4 mm hole geometrical specification (rule D).

Figure A.17 — Simultaneous specifications with MMR

Drawing indication



The interpretation of the first location specification (SZ CZ) is based on the following rules and definitions given in this document.

i) The extracted features of the six toleranced features shall not violate the MMVCs, which have the diameters MMVS = 19,9 – 0,1 = 19,8 mm [see rule C, 3.9 and 4.1.2].

ii) The extracted features of the six toleranced features shall have everywhere a local diameter smaller than LMS = 20,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 19,9 mm [see rule A 2), 3.4 and 3.5].

iii) MMVCs are constrained in orientation and location according to rule D and the rules of ISO 5458. Three MMVCs are constrained to have axes parallel between them, to lie on a cylinder of radius 40 mm, to be equally disposed on the cylinder R40 with angles of 120°. The axis of the cylinder R40 is also constrained to be aligned with the datum. Three other MMVCs are also constrained to have axes parallel between them, to lie on a cylinder of radius 40 mm, to be equally disposed on the cylinder R40 with angles of 120°. The axis of the cylinder R40 is also constrained to be aligned with the datum. The first group of three MMVCs is not constrained to be at the same angular position as the second group of three MMVCs.

iv) The datum is defined according to the rules of ISO 5459.

The interpretation of the second location specification (CZ CZ) is based on the following rules and definitions given in this document.

a) The extracted features of the six toleranced features shall not violate the MMVCs, which have the diameter MMVS = 19,9 – 0,3 = 19,6 mm [see rule A 2), 3.4 and 3.5].

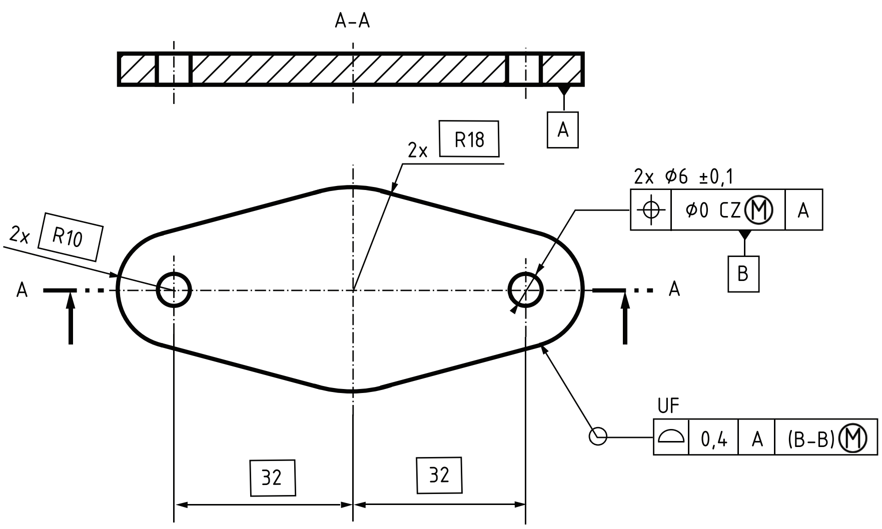
b) The extracted features of the six toleranced features shall have everywhere a local diameter smaller than LMS = 20,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 19,9 mm [see rule A 2), 3.4 and 3.5].

c) MMVCs are constrained in orientation and location according to rule D and the rules of ISO 5458. Three MMVCs are constrained to have axes parallel between them, to lie on a cylinder of radius 40 mm, to be equally disposed on the cylinder R40 with angles of 120°. The axis of the cylinder R40 is also constrained to be aligned with the datum. Three other MMVCs are also constrained to have axes parallel between them, to lie on a cylinder of radius 40 mm, to be equally disposed on the cylinder R40 with angles of 120°. The axis of the cylinder R40 is also constrained to be aligned with the datum. The first group of three MMVCs is constrained to be at the same angular position as the second group of three MMVCs.

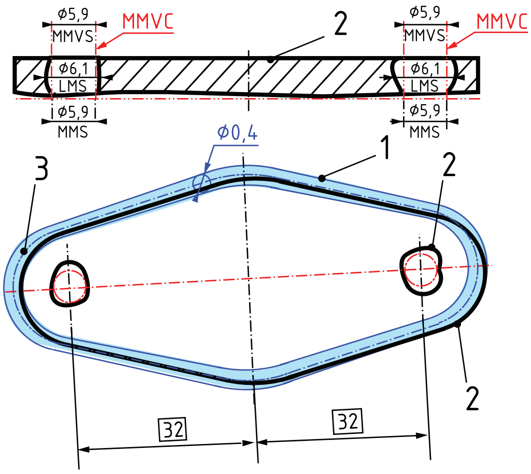
d) The datum is defined according to the rules of ISO 5459.

Figure A.18 — Specifications for patterns with MMR

Drawing indication



Interpretation



The interpretation of the location specification is based on the following rules and definitions given in this document.

i) The extracted features of the two toleranced features (holes) shall not violate the MMVCs, which have the diameters MMVS = 5,9 mm [see rule C, 3.9 and 4.1.2).

ii) The extracted features of the two toleranced features (holes) shall have everywhere a local diameter smaller than LMS = 6,1 mm [see rule B 2), 3.6 and 3.7] and larger than MMS = 5,9 mm [see rule A 2), 3.4 and 3.5].

iii) The two MMVCs are in perfect orientation from the datum and in perfect orientation and location relative to each other [see rule D and 3.9, Note 2 to entry].

The interpretation of the surface profile specification is based on the following rules and definitions given in this document and ISO 1101.

a) The tolerance zone is defined by an infinite number of spheres whose diameter is equal to 0,4 mm, the centres of which are situated on the theoretically exact feature (TEF). The TEF is in perfect position and orientation from the datum (see ISO 1101).

b) The extracted features of the datum shall not violate the MMVCs, which have the diameter MMVS = 5,9 mm [see rules E and G, 3.8 and 3.9]. MMVCs are in perfect orientation and location relative to each other.

Key

1 surface profile tolerance zone

2 real part

3 theoretically exact feature (TEF)

Figure A.19 — Specification with MMR on datum only

Annex B
(informative)

Former practice

B.1 General

This annex describes former practices that have been omitted and are no longer used. Therefore, they are not an integral part of this document and should be used for information only.

B.2 Use of CZ and SZ symbols

In the case of several toleranced features controlled by the same tolerance indication, the MMVCs or LMVCs were considered in theoretically exact orientation and location relative to each other (see rule D and ISO 2692:2014, 4.2.1 NOTE 4).

To avoid ambiguity and to remain in line with other GPS standards, CZ or SZ symbols shall be indicated according to this document (see 4.1.4). Table B.1 shows the evolution of indications when requirements apply to several toleranced features.

Table B.1 — Evolution of indication for patterns

	ISO 2692:2014	**ISO 2692:2021**
Combined	2x | | ⌀0,2 Ⓜ |	2x | | ⌀0,2 CZ Ⓜ |
Separate	Not applicable	2x | | ⌀0,2 SZ Ⓜ |

B.3 Use of datum feature indicator

In ISO 2692:2014 rule G, the datum feature indicator was required to be directly connected to that geometrical tolerance indicator from which MMVC of the datum feature is controlled (see ISO 5459:2011, Rule 1, dash 2).

In this document it is only a recommendation.

In ISO 2692:2014 rule N, the datum feature indicator was required to be directly connected to that geometrical tolerance indicator from which LMVC of the datum feature is controlled (see ISO 5459:2011, Rule 1, dash 2).

In this document it is only a recommendation.

Annex C
(informative)

Concept diagram

Figure C.1 shows the concepts for the case where indirect determination of virtual size is selected (see 4.1.2).

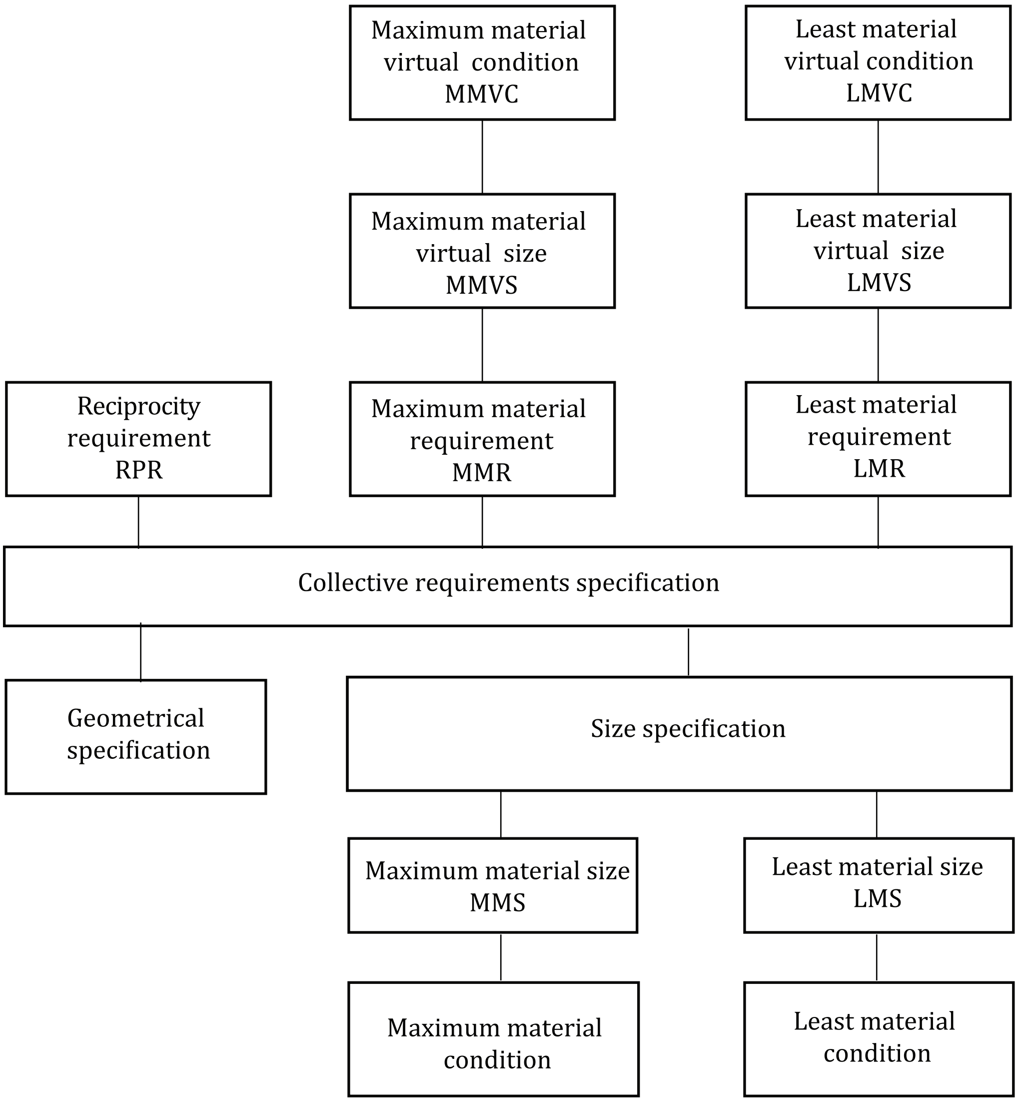


Figure C.1 — Concept diagram for terms and concepts related to MMR and LMR with indirect virtual size determination

Figure C.2 shows the concepts for the case where direct indication of virtual size is selected (see 4.1.3).

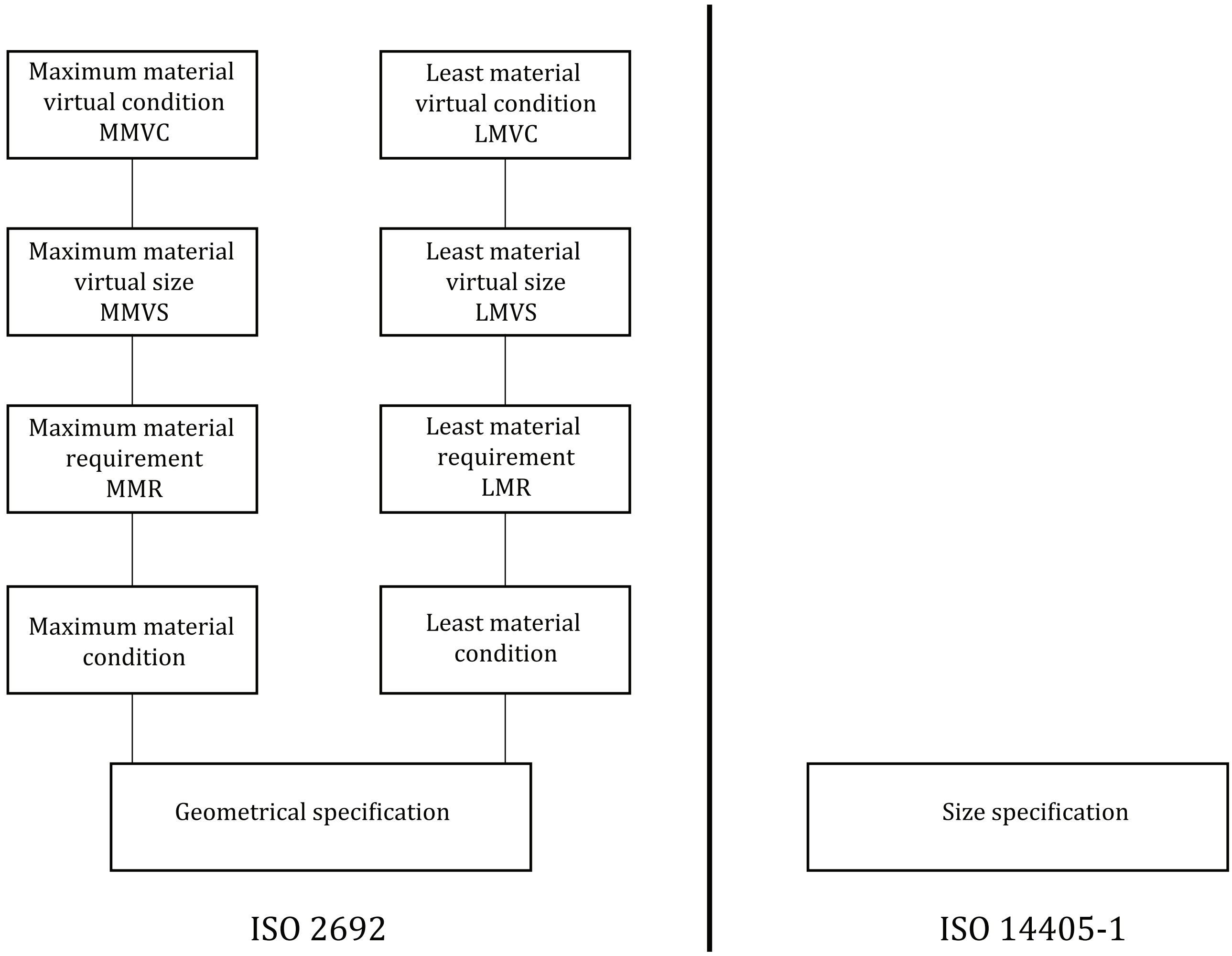


Figure C.2 — Concept diagram for terms and concepts related to MMR and LMR with direct virtual size indication

Annex D
(informative)

Use of symbols for geometrical characteristics with Ⓜ or Ⓛ

D.1 General

A geometrical specification with MMR or LMR is always related to a derived feature of linear size of cylindrical type or of two parallel opposite planes type. For derived feature indication, see ISO 1101:2017, Clause 6.

A common property of the symbols that are possible to use with MMR or LMR (see D.2) is that they can appear in geometrical specification applying to a derived feature which is:

— a median line of a feature of linear size of cylindrical type; or

— a median surface of a feature of linear size of two parallel opposite planes type.

D.2 Use of symbols for geometrical characteristics

Table D.1 describes whether or not it is possible to use MMR or LMR with the various symbols for geometrical characteristics.

Table D.1 — Use of symbols for geometric characteristics

Specification	Characteristic	Symbol	Is it possible to use the symbol with Ⓜ or Ⓛ ?
Form	Straightness	⏤	Yes
	Flatness	⏥	Yes
	Roundness	○	No
	Cylindricity	⌭	No
	Line profile	⌒ [a]	No
	Surface profile	⌓ [b]	No
Orientation	Parallelism	∥	Yes
	Perpendicularity	⊥	Yes
	Angularity	∠	Yes
	Line profile	⌒ [c]	No
	Surface profile	⌓ [d]	No

[a] It is possible to use line profile symbol when equivalent to the straightness.

[b] It is possible to use surface profile symbol when equivalent to the flatness.

[c] It is possible to use line profile symbol when equivalent to the orientation of a nominal straight line.

[d] It is possible to use surface profile symbol when equivalent to the orientation of a nominal plane.

[e] It is possible to use line profile symbol when equivalent to the location of a nominal straight line.

[f] It is possible to use surface profile symbol when equivalent to the location of a nominal plane.

Table D.1 *(continued)*

Specification	Characteristic	Symbol	Is it possible to use the symbol with Ⓜ or Ⓛ ?
Location	Position	⌖	Yes
	Coaxiality	◎	Yes
	Symmetry	⌯	Yes
	Line profile	⌒ [e]	No
	Surface profile	⌓ [f]	No
Run-out	Circular run-out	↗	No
	Total run-out	⌰	No

[a] It is possible to use line profile symbol when equivalent to the straightness.

[b] It is possible to use surface profile symbol when equivalent to the flatness.

[c] It is possible to use line profile symbol when equivalent to the orientation of a nominal straight line.

[d] It is possible to use surface profile symbol when equivalent to the orientation of a nominal plane.

[e] It is possible to use line profile symbol when equivalent to the location of a nominal straight line.

[f] It is possible to use surface profile symbol when equivalent to the location of a nominal plane.

Annex E
(informative)

Relation to the GPS matrix model

E.1 General

For full details about the GPS matrix model, see ISO 14638.

E.2 Information about the document and its use

This document defines the MMR, the LMR and the RPR applied to features of linear size of type cylinder or two parallel opposite planes.

E.3 Position in the GPS matrix model

This document is a general GPS standard that influences chain links A, B and C of the chains of standards on size, form, orientation and location in the GPS matrix model, as graphically illustrated in Table E.1.

Table E.1 — Position in the GPS matrix model

	Chain links						
	A	B	C	D	E	F	G
	Symbols and indications	**Feature requirements**	**Feature properties**	**Conformance and non-conformance**	**Measurement**	**Measurement equipment**	**Calibration**
Size	•	•	•				
Distance							
Form	•	•	•				
Orientation	•	•	•				
Location	•	•	•				
Run-out							
Profile surface texture							
Areal surface texture							
Surface imperfections							

E.4 Related International Standards

The related International Standards are those of the chains of standards indicated in Table E.1.

Bibliography

[1] ISO 286-1, *Geometrical product specifications (GPS) — ISO code system for tolerances on linear sizes — Part 1: Basis of tolerances, deviations and fits*

[3] ISO 7083, *Technical product documentation — Symbols used in technical product documentation — Proportions and dimensions*

[4] ISO 8015, *Geometrical product specifications (GPS) — Fundamentals — Concepts, principles and rules*

[5] ISO 14253-1, *Geometrical product specifications (GPS) — Inspection by measurement of workpieces and measuring equipment — Part 1: Decision rules for verifying conformity or nonconformity with specifications*

[6] ISO 14638, *Geometrical product specifications (GPS) — Matrix model*

[7] ISO 22432, *Geometrical product specifications (GPS) — Features utilized in specification and verification*

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